Supplementary Material

Incentives, cooperation and accountability in Brazil's "Chapecó Consensus"

This Supplementary Material complements the article in eight units, gathered in a single document. Each unit addresses one theme and can be read on its own, and the article cites each unit at the point where it provides details. The first unit expands the institutional background, the next two concern the model, the following four concern the data, the calibration and the results that the article summarizes, and the last one describes the tests of the engine and how the results can be reproduced.

- Supplementary Material 01 – Institutional Background gives the legal history and scale of individual amendments, the evidence on what they buy, the episodes in which a channel of discretion was closed, the territorial institutions on which the proposal could build, the completion of works and two foreign precedents (Sections S1.1–S1.6).
- Supplementary Material 02 – Model Specification states the full model: time, agents and money, the value of local money, the Chamber, the twelve stages of the county game, the reference designs, the managerial law, the measure of value and the computation (Sections S2.1–S2.8, Table S1 and equations S1–S10).
- Supplementary Material 03 – Proofs and Numerical Checks restates and proves Propositions 1–7 of the article and Supplementary Propositions 8 and 9, each with the idea of the proof and a proof in numbered steps, and reports the brute-force checks of Propositions 1–7 (Sections S3.1–S3.10, Table S2 and equations S11–S41).
- Supplementary Material 04 – Data and Calibration documents the data, the construction of the variables and of the status quo, the calibration of the parameters and the checks of the calibrated model against data (Sections S4.1–S4.9, Tables S3–S5 and Figure S1).
- Supplementary Material 05 – Additional Results and Robustness Checks reports the results of every design, the gains and losses of each municipality, the national accounting of the per-capita rule and the robustness checks (Sections S5.1–S5.4, Tables S6–S8 and Figures S2 and S3).
- Supplementary Material 06 – Sensitivity Analysis and Scenario Discovery describes the screening of the parameters, the grouped Sobol' indices and the classification trees that search for the conditions under which each result holds (Sections S6.1 and S6.2, Table S9 and Figure S4).
- Supplementary Material 07 – Colleges of Mayors and the Electoral Connection describes the real colleges of mayors elected in 2024 and the evidence on deputies' votes and transfers (Sections S7.1 and S7.2, Table S10 and Figures S5 and S6).
- Supplementary Material 08 – Tests and Replication describes the tests of the engine and the replication package (Sections S8.1 and S8.2).

Sections, equations, tables and figures numbered with the prefix S belong to this Supplementary Material, and the first digit of a section number identifies its unit (Section S4.2, for example, is in Supplementary Material 04). Sections, equations, propositions, tables and figures without the prefix refer to the main article.

Supplementary Material 01 – Institutional Background

This unit expands Section 2 of the article, which summarizes the institutions that the proposal would change. It gives the legal history and the scale of individual amendments (Section S1.1), the evidence on what amendments buy (Section S1.2), the episodes in which a channel of discretion was closed (Section S1.3), the territorial institutions on which the proposal could build (Section S1.4), the evidence on the completion of works (Section S1.5) and two foreign reforms close to the two halves of the proposal (Section S1.6).

S1.1. Individual amendments

Members of Congress may amend the annual budget bill, and since 2015 the amendments of individual members have had a guaranteed place in it. Constitutional Amendment 86 fixed individual amendments at up to 1.2% of the projected net current revenue of the federal government, with half earmarked for health, and made their execution compulsory. Two later amendments widened this space. Constitutional Amendment 105 (2019) created the "special transfer", which sends the money directly to the receiving state or municipality without an agreement on its use, and Constitutional Amendment 126 (2022) raised the ceiling to 2% of the previous year's net current revenue, 1.55% for deputies and 0.45% for senators. Under these rules, in 2025 each deputy controlled R$ 37.2 million and each senator R$ 68.5 million, about R$ 24.6 billion in total. Adding the amendments of state delegations ("bench" amendments) and of standing committees, the 2025 budget carried about R$ 50 billion in amendments (Agência Senado, 2025). Over a longer period, committed amendments rose from R$ 11.3 billion in 2014 to R$ 47.1 billion in 2025 at constant prices, according to a 2026 study by the Institute for Applied Economic Research (Ipea), as reported in the press (Pavan, 2026).

Two features distinguish the Brazilian case. The first is scale. Amendments paid represented about 24% of federal discretionary spending in 2024 (a projection based on mid-year data). Among the countries compared by Tollini and Mendes (2024), the corresponding shares were below 1% in France, Italy, Portugal and Spain, 4.4% in Korea and 9% in Germany; in the United States, amendments directing funds to states and localities are capped at 1% of discretionary spending. The second feature is that the legislator chooses the recipient and the Executive must comply. Congress has repeatedly tried to expand this discretion, and the Supreme Court has restrained it. In December 2022, the Court declared unconstitutional the "rapporteur amendments", a mechanism that let the budget rapporteur distribute funds at the request of unidentified legislators (InfoMoney, 2022). In August 2024, it suspended the execution of mandatory amendments until transparency rules were adopted (Gazeta do Povo, 2024), which led to Complementary Law 210/2024.

The special transfers have a further property that is useful for the article: they make the geography of part of the amendment money observable. In 2025, the 15 federal deputies of Santa Catarina who used them planned R$ 243 million in special transfers to the state's municipalities, 41% of the 16 deputies' combined quota (Section 5.1 of the article). This is the allocation that the reform would replace, and Section 6.3 of the article shows that it is far from neutral with respect to the size and the party of the receiving municipality.

S1.2. What amendments buy

Three bodies of evidence bear on the proposal. The first concerns the electoral connection, that is, the link between the money that deputies direct and the votes they receive. Under open-list proportional representation, deputies cultivate geographically concentrated vote bases and target amendments to them (Ames, 1995). Samuels (2002) reports that 65% of the incumbents elected in 1990 drew at least 20% of their personal votes from a single municipality. He finds no direct effect of amendments on reelection and argues that pork pays indirectly, by attracting campaign contributions. In a related study, Samuels (2003) shows that the careers of Brazilian deputies point toward state and municipal executive offices, which gives brokerage a payoff beyond reelection to the Chamber.

Later studies measure the targeting more directly. Firpo et al. (2015) show that deputies allocate amendments to municipalities where they are electorally strong. Baião and Couto (2017) estimate a positive elasticity of a deputy's municipal vote with respect to amendments, larger where the mayor belongs to the deputy's party, and Ten Cate (2023) reports that 22.5% of individual amendments in 2015–2019 went to municipalities governed by the author's party. As for the effects of the money on the municipalities that receive it, Bertholini et al. (2018) find lower infant mortality and better formal-employment indicators in municipalities that received amendments, with diminishing marginal effects. The same targeting appears in Santa Catarina: the special transfers that deputies sent in 2025 followed their 2022 votes and their party's mayors, and they are associated with larger vote shares in 2026 (Section 5.1 of the article and Supplementary Material 07, Section S7.2).

The second body of evidence concerns the exchange between the Executive and the legislature. Presidents have used amendments as a currency to obtain support for their agenda (Pereira & Mueller, 2004; Alston & Mueller, 2006; Raile et al., 2011), although Limongi and Figueiredo (2005) argue that the execution of amendments follows coalition membership and not individual votes. Studies of the president's "toolbox" also document several currencies that can substitute for one another, such as cabinet posts and appointments (Raile et al., 2011), and models of vote buying in legislatures treat the currency as fungible (Groseclose & Snyder, 1996). If currencies substitute for one another, abolishing one of them may lead to its replacement by another, a possibility that Section S1.3 examines. In the same vein, Brollo and Nannicini (2012) show that discretionary federal transfers favor municipalities governed by the president's coalition by about one-third in pre-election years.

The third body of evidence concerns brokers. Mayors are important vote brokers for federal deputies, but they are disloyal agents. Before a 2007 ruling against party switching, 39% of the mayoral candidates who ran in two consecutive elections switched parties, and 31% did so afterward (Novaes, 2018). In Santa Catarina, 35% of the mayors reelected in 2024 had changed party since 2020, and 18 of the 29 who switched joined the party of the state governor (Supplementary Material 04, Section S4.3). Mayors also pursue discretionary money as an electoral asset: mayors who can be reelected are 14 percentage points more likely to obtain a federal

matching grant in the election year (Ferraz & Finan, 2011). Because the proposal would make mayors the electors of the condes, their party loyalty becomes part of the reform itself, and the model represents it through party loyalty in the college vote and party switching before municipal elections (Section 4.4 of the article).

S1.3. When a channel of discretion closes

Whether brokerage disappears or relocates when one of its channels is closed is the question on which the Chamber half of the proposal turns, and Brazil has a recent precedent. After the Supreme Court ruling of December 2022, the R$ 19.4 billion reserved for rapporteur amendments in the 2023 budget bill was divided roughly in halves between a larger individual-amendment quota and the ministries' regular appropriations (Congresso em Foco, 2022). In the following years, committee amendments, which had amounted to only R$ 0.3 billion in the 2022 budget, rose to R$ 6.9 billion in 2023 and R$ 14.2 billion in 2024. Complementary Law 210/2024 capped them at R$ 11.5 billion in 2025, but, because discretionary appropriations were also channeled through committee amendments, the 2025 total reached R$ 19 billion (Ronconi et al., 2025).

Closing a channel of campaign money had a similar effect. When corporate donations were banned in 2015, Congress compensated parties with more public funds, a fourfold increase according to Avis et al. (2022), and mayoral candidates self-financed 40% of their campaign spending in 2016, against 25% in 2012 (Avis et al., 2022). The episode of the rapporteur amendments does not show what would happen if all parliamentary amendments were abolished. Even so, it suggests that the political demand for discretion is met through the channels that remain, which is why the model treats the share of the abolished money that reappears in other discretionary channels as an uncertain parameter (Section 4.3 of the article and Supplementary Material 04, Section S4.9).

S1.4. Counties, consortia and regional bodies

The second half of the proposal can build on existing territorial units. The SUS divides the country into about 450 health regions (456 in March 2023), grouped into macro-regions (Conselho Nacional de Secretários de Saúde [CONASS], 2023). In Santa Catarina, Deliberation 38/CIB/2024 of the state's bipartite health committee defines 17 health regions and 8 macro-regions, which are, respectively, the counties of the base design and those of a robustness check. Municipalities also cooperate through consortia. Federal Law 11.107/2005 regulates public consortia among federal entities, which finance common services through apportionment contracts and may exclude members that fail to budget their share; 69.2% of Brazilian municipalities belonged to at least one public consortium in 2019 (Agência IBGE Notícias, 2020). In Santa Catarina, the state federation of municipalities (FECAM) brings together about twenty regional associations of municipalities, which have supported the formation of consortia (Abrucio et al., 2013).

The state also offers a precedent closer to the proposal. In 2003, Santa Catarina created 29 Regional Development Secretariats, later 36, with regional councils that included mayors. The secretariats were converted

into regional agencies in 2015 and abolished in 2019, and Filippim and Abrucio (2010) argue that their councils drifted toward the distribution of local works. This precedent points to a risk that the model represents explicitly: a regional body in which mayors decide may favor local works over regional ones (Section 4.4 of the article).

Evidence from regional bodies elsewhere is mixed. In US metropolitan planning organizations, boards with a larger share of elected officials allocate less to regional projects (Gerber & Gibson, 2009). The cost savings expected from intermunicipal cooperation are not found systematically (Bel & Warner, 2015; Bel & Sebő, 2021), and the institutional collective action framework predicts that cooperation becomes harder as partners grow more heterogeneous and gains more diffuse (Feiock, 2013). On the other hand, in Brazilian water and sanitation, municipal companies almost doubled their investment after a legal reform removed the threat that state companies would take over their systems (Kresch, 2020).

S1.5. The completion of works

Completion is the weak link. In 2019, the Federal Court of Accounts (TCU) found that 14,403 of the 38,412 federally funded works it reviewed (37.5%) were paralyzed or unfinished. The main causes were deficient basic designs, insufficient counterpart funds from the co-responsible states or municipalities, and these governments' difficulty in managing the federal money they received (Tribunal de Contas da União [TCU], 2019). In 2023, the Court counted 8,603 paralyzed works among 21,007 (41%), and 38.9% in Santa Catarina (TCU, 2023). Turnover in municipal government adds to the problem: in Brazilian municipalities, a change of the party in power disrupts the local bureaucracy and lowers the quality of municipal services, with test scores in municipal schools 0.05–0.08 standard deviations lower (Akhtari et al., 2022). A regional project depends on all of these conditions, namely the technical capacity to design it, the co-financing of several municipalities and continuity across terms, and for this reason completion, co-financing and continuity are explicit parts of the model (Section 4.4 of the article and Supplementary Material 02, Section S2.4).

S1.6. Two foreign precedents

Two foreign reforms are close to the two halves of the proposal. France abolished the "parliamentary reserve", the practice of funding operations proposed by individual members of Parliament, in 2017 (Organic Law 2017-1338), and a 2023 Senate bill to restore it has not been enacted. Portugal began electing the presidents of its five regional coordination and development commissions indirectly in October 2020, through regional electoral colleges of mayors, councilors and municipal assembly members, more than 10,000 local elected officials in all (Decree-Law 27/2020, as amended by Law 37/2020; Mendonça, 2020). In four of the five regions there was a single candidate (Mendonça, 2020).

Supplementary Material 02 – Model Specification

This unit states the model of Section 4 of the article in full. The article presents the main elements of the model and its propositions; here, every stage is specified with the functional forms and parameters that the computation uses, so that the model can be checked and reproduced. The unit proceeds from the general to the particular. Section S2.1 sets out time, agents and money, and Section S2.2 defines the value of local money. Section S2.3 describes the Chamber and Section S2.4 the twelve stages of the county game. Sections S2.5 and S2.6 describe the reference designs and the managerial law, Section S2.7 defines value and Section S2.8 explains the computation. Table S1, at the end of the unit, collects the notation.

S2.1. Time, agents and money

The model is annual and is solved term by term. There are three conde terms, $\tau = 1,2,3$ (2027–2030, 2031–2034 and 2035–2038). Each term has two halves of two years, separated by the municipal election held in October of the term's second year (2028, 2032 and 2036). This calendar determines who votes for the conde: the college that elects the conde in 2027 is made of the mayors elected in October 2024 (real data), while the colleges of 2031 and 2035 are made of the mayors produced by the model's municipal elections. The agents are the 16 federal deputies of Santa Catarina and the list-makers of their eight parties (a federation counts as one), the Executive (which acts through the relocation of money), the 295 mayors, the 17 condes and the Union, which sets and enforces the rules.

The money is fixed, so that every design distributes the same amount, M, equal to R$ 595.2 million a year, the sum over municipalities of the status-quo allocation x_m^{SQ} (Supplementary Material 04 – Data and Calibration, Section S4.2). In the status quo, the money is allocated with the 2025 geography in every year. Under a reform that abolishes amendments, a share η reappears in discretionary channels that deputies still broker (Section 4.3 of the article) and is allocated, as today, with the status-quo geography, $\eta\, x_m^{SQ}$ a year; the rest, $(1-\eta)M$ a year, goes to the counties. County j receives $B_{j\tau} = 4(1-\eta)M\,\varsigma_{j\tau}$ in term τ, where $\varsigma_{j\tau}$ is its share of the county money. This share is the county's population share, except in R10 and R16, where it is the county's population times its mean need, normalized. It is then multiplied by $1 - s_T\Phi_j$ if the county failed its targets in the previous term and the cut applies, and by $1 + 1/3$ if its conde is aligned with the federal government in R11, and the shares are renormalized to sum to one. A share c_{adm} of the county money pays for the county administration; the rest, $\overline{B}_{j\tau}$, is the money the conde allocates.

S2.2. The value of local money

Every real that reaches a municipality, whether as an amendment, as relocated money or as a county transfer, is valued in the same way. A real transferred to municipality m is worth

$$v_m = (1 - d_m)\,[\phi + (1 - \phi)(q_m + (1 - q_m)\,v_{inc})], \tag{S1}$$

where $\phi = 0.3$ is the share spent on current items, v_{inc} the value per real of an unfinished work and q_m the probability that an investment is completed within the term,

$$q_m = \min\{1,\ \bar{q}\,(\kappa_m/\bar{\kappa})^{\nu}\left(1 + \mathbb{1}_{LRG}\,g_{TA}(1 - \kappa_m)\right)\}, \tag{S2}$$

with $\kappa_m \in (0,1)$ the municipality's technical capacity (Supplementary Material 04, Section S4.2) and $\bar{\kappa}$ its mean. Equations (S1) and (S2) extend equation (1) of the article in one respect: with the managerial law, technical assistance multiplies the completion probability by $1 + g_{TA}(1 - \kappa_m)$, up to the cap of one, so that the proportional gain is largest where capacity is lowest. Diversion is $d_m = f_{SC}\,\bar{d}_m\,I$, with $\bar{d}_m = 0.055$ for a mayor who can be re-elected and 0.074 for a term-limited mayor (Ferraz & Finan, 2011), $f_{SC} \in [0.5,1]$ a factor for Santa Catarina and I the inspection factor of the managerial law ($I = 1$ without the law; Section S2.6). Because the term-limit status of a mayor changes at each municipal election, v_m is computed separately for each half of the term.

S2.3. The Chamber

The Chamber module has three parts: who is elected, how the elected divide their effort, and what the Chamber produces. The first two parts follow Propositions 1 and 2 of the article, and the third combines them into a measure of output.

S2.3.1. Selection

Party p holds S_p seats, the 2026 result with each federation's seats given to its largest party: PL 7, PT 3, MDB 1, Novo 1, PSD 1, PP 1, Podemos 1 and Republicanos 1 (Supplementary Material 04, Section S4.2). The party chooses its deputies among $N_p = \lceil k_{asp} S_p \rceil$ aspirants, whose legislative and brokerage skills (λ_L, λ_B) are standard bivariate normal with correlation ρ. Under open lists, the elected are the S_p aspirants with the highest $\lambda_L + t_0\lambda_B + \sigma_\varepsilon\varepsilon$; under closed lists, they are the S_p aspirants with the highest $\lambda_L + X_p t_0 \lambda_B + \sigma_\nu \nu$, where ε and ν are independent standard normal. The expected skills of the elected follow from Proposition 1.

S2.3.2. Effort

A deputy of party p chooses brokerage effort e_B to maximize $W_B \ln e_B + W_L \ln(1 - e_B)$. In the status quo, $W_B/(W_B + W_L) = s_0$. Under closed lists, the weights become $\pi\psi_p W_B$ and κW_L, where

$$\psi_p = \eta + \chi(1-\eta)\,\bar{s}_p \qquad \text{(S3)}$$

is the brokerage technology that survives the abolition of amendments ($\psi_p = 1$ if amendments are kept, as in CL), χ the responsiveness of a co-partisan conde to deputies' requests and $\bar{s}_p$ the share of county money controlled by party p's condes, averaged over the three terms. Hence $X_p = \pi\psi_p/\kappa$ and $e_{B,p} = X_p s_0/(X_p s_0 + 1 - s_0)$ (Proposition 2). The term $\bar{s}_p$ is the channel through which the county game feeds back into the Chamber.

S2.3.3. Output

The legislative output of the Chamber is $Y = \sum_p w_p\,(1 + \gamma_L \bar{\lambda}_{L,p})(1 - e_{B,p})$, where w_p is party p's seat share and $\bar{\lambda}_{L,p}$ the expected legislative skill of its deputies, reported relative to the status quo. The measure is decomposed in two ways: holding effort at its status-quo level gives the selection effect, and holding skill at its status-quo level gives the effort effect.

S2.4. The county game

Each term has twelve stages, solved for every county and world. Stages 1 to 8 (Sections S2.4.1–S2.4.8) describe the election of the conde, the menu of projects, the technologies of completion and capture, the horizons and the cooperation of mayors, the plan and the conde's effort. Stages 9 to 12 (Sections S2.4.9–S2.4.12) describe what happens once these choices are made: the realization of outcomes, the pre-election tilt, the municipal elections and, at the end of the term, targets and sanctions.

S2.4.1. Election of the conde

Each party with at least one mayor in the county fields a candidate. The incumbent's party fields the incumbent if the incumbent may run, that is, if the incumbent is not barred by a missed target and, in R1, has not completed one term. Mayor m votes for party p with probability proportional to $\exp(\varphi_P \mathbb{1}\{p_m = p\} + \rho_R\, g_m\, \mathbb{1}\{p = p_{inc},\ \text{incumbent runs}\})$, where g_m is the value municipality m received in the previous term (transfers and projects, net of its co-financing) relative to the value of a per-capita share of the county money, minus one. The candidate with the most votes wins (votes weighted by population in R6), and ties are broken at random. A new conde draws a weight $\theta \sim U[\theta_{lo}, 1]$ on the value of the money relative to mayors' valuation and a competence $a_c \sim N(0,1)$; a reappointed conde keeps both. The tenure of the conde's party counts the consecutive terms in which the party has held the office. With a technical conde (R12), there is no vote: the conde has no party and $\theta = 1$, its competence is drawn from the same distribution as that of political condes, and it is reappointed if and only if the county meets its targets.

S2.4.2. The menu

The county has $K = 6$ candidate projects. Project k costs $C_k = \bar{s}\,\bar{B}_{j\tau}\,G_k / \sum_l G_l$, with $G_k \sim$ Gamma(2,1), and is worth $V_k = (1 + \sigma_k)C_k$ if completed, with $\sigma_k = \max\{\bar{\sigma} + sd_\sigma z_k, -0.5\}$ and $z_k \sim N(0,1)$. The host is the county's largest city with probability p_{polo} and otherwise a member drawn with probability proportional to the square root of its population. Municipality m's share of the benefits is $w_{mk} \propto pop_m \exp(-dist(m, h_k)/D_0)$, so that benefits decay with distance from the host. The municipalities with $w_{mk} \geq 0.02$, and the host, participate. They pay the co-financing share $\chi_{cof} = 0.3$ of the cost in proportion to $\tilde{w}_{mk}$, their benefit shares renormalized among participants, and the county pays $C_k^U = (1 - \chi_{cof})C_k$. Without co-financing (R5), the county pays the whole cost.

S2.4.3. Completion technology

With supervision effort e, the probability that project k is completed within the term is

$$q_k = Q(e)\,[1 - \omega_h(1 - h_k)]\,[(1 - \chi_{cof}) + \chi_{cof} A_k]^{\nu_c}, \tag{S4}$$

where the technical term is

$$Q(e) = \min\{1,\ Q_0(1 + \alpha_e e)\}, \tag{S5}$$

with

$$Q_0 = \bar{q}\,(1 - \phi_{cx})\left(\kappa_j/\bar{\kappa}\right)^{\nu}(1 + \alpha_a a_c)\left(1 + \mathbb{1}_{LRG}\,g_{TA}(1 - \kappa_j)\right). \tag{S6}$$

In these equations, $h_k = 1$ if the host cooperates, A_k is the share of the co-financing actually paid and $\kappa_j = \bar{\kappa}_j + g_{pool}(1 - \bar{\kappa}_j)$ is the capacity of the county, the population-weighted mean of its members' capacity raised by a pooled technical team. The technical term $Q(e)$ thus combines the completion of transfer-funded works, the penalty of regional complexity, the county's capacity, the conde's competence and effort and, with the managerial law, technical assistance.

S2.4.4. Diversion and capture

County procurement loses a share

$$d_c = d_{c0}\,(1 - D_c)\,\big(1 + g_{ten}(\text{tenure} - 1)\big)\,I, \tag{S7}$$

where the discipline of a contestable reappointment is

$$D_c = D_{max}\,\mathbb{1}\{\text{reappointment possible}\}\max\{0,\ 1 - \text{sup}/0.55\} \tag{S8}$$

and sup is the share of the college held by the conde's party. In words, a contestable reappointment disciplines the conde, as re-election disciplines mayors (Ferraz & Finan, 2011), but the discipline weakens as the conde's party grows in the college and vanishes when the party holds 55% of it; capture also rises with the party's tenure.

S2.4.5. Horizons

A participating mayor values future projects with the discount factor $\delta_m = \beta_y^4\,\pi_m^{stay}\,\pi_j^{cont}$, which combines the four years of a term with two probabilities. The probability of being in office next term is $\pi_m^{stay} = r_{run}p_{win}$ for a mayor who may run again and zero for a term-limited mayor. The continuity of the conde's office is $\pi_j^{cont} = p_j^{next}\,[f + \kappa_P(1 - f)]$, where p_j^{next} is the probability that the conde's party wins the next college, κ_P the share of commitments a co-partisan successor honors and f the probability that the conde may stay. This probability is zero in R1 and one without the bar. With the bar, it is the probability that the county meets its targets as mayors expect it from the county's capacity and its delivery in the previous term, $f_j = \Phi\left(\left(\sqrt{w_c}\,cap_j + \sqrt{w_p}\,perf_{j,\tau-1} - \bar{z}\right)/\sqrt{w_n}\right)$, with $perf_{j,0} = 0$ (Section S2.4.12 defines the score). The probability $p_j^{next} = \Phi(\mu_L/sd_L)$ is computed from the party's expected lead in the next college: $\mu_L = (n_c - n_{riv})\,\rho_{ret}$, where n_c and n_{riv} are the shares of the college held by the conde's party and by its strongest rival, $\rho_{ret} = r_{run}p_{win}(1 - h_{sw}) + (1 - r_{run}p_{win})r_{keep}$ is the probability that a seat stays with its party, and $sd_L = (1/2\sum_m \omega_m^2)^{1/2}$, with ω_m the vote weights. For a technical conde, $p_j^{next} = 1$ and $f = f_j$.

S2.4.6. Cooperation

Given the plan, each participant decides whether to pay its co-financing share, and the host whether to provide land, permits and local execution, after observing a private cost shock $\epsilon_{mk} \sim \text{Logistic}(0, \varsigma\,stake_{mk})$. The equilibrium is the fixed point of Proposition 3, computed by iterating best responses six times from $p_{mk} = 0.7$. A non-payer loses a share ϵ_x of the project's benefits, a share raised to $\epsilon_x + (1 - \epsilon_x)u_{contract}$ by binding contracts in R3, and, with probability δ_m, the next term's projects. A mayor's credit for regional projects is $\tilde{\gamma}_m = \gamma - \zeta_{opp}\mathbb{1}\{p_m \neq p_c\}$, so that mayors of a party other than the conde's receive less credit. With co-

branding (R4), the credit is $\gamma + \gamma_{cb}$, and the conde's private credit falls to $\kappa_{cred}(1 - \gamma_{cb})$. The stake is $(1 + \ell)\chi_{cof} C_k \tilde{w}_{mk}$, plus $c_{host} C_k$ for the host.

S2.4.7. The plan and its approval

The conde values project k at

$$U_k = \theta\left\{(1 - d_c)\left[q_k V_k + (1 - q_k)\, U_k^{unf}\right] - \bar{v}_j\, spent_k\right\} + (1 - \theta)\left\{\frac{\sum_m \left[(1 + \tilde{\gamma}_m) w_{mk} V_k q_k - p_{mk} stake_{mk}\right]}{1 + \ell} - (1 - \kappa_{cred})\, C_k^U\right\}, \quad \text{(S9)}$$

where $spent_k = C_k^U + \chi_{cof} C_k A_k$, $\bar{v}_j$ is the population-weighted value of local money in the county and $U_k^{unf} = (1 - \bar{a})\left(q_k \beta_d V_k + (1 - q_k) v_{inc} spent_k\right) + \bar{a}\, v_{inc} spent_k$ is the expected value of an unfinished project, which is continued next term or abandoned with probability $\bar{a}$ (the probability that the office changes hands, weighted by a_{ab} and a_{same}). The first brace of equation (S9) is the social value of the project net of the local money it displaces; the second is the mayors' valuation of it, net of their expected stakes (co-financing and, for the host, the local cost), minus the county money it uses, which the conde's private credit partly offsets. The conde ranks projects by U_k / C_k^U and includes those with $U_k > 0$ while their county cost stays below 90% of the county money; the rest is transferred to municipalities in proportion to population.

The plan needs the approval of a majority of mayors (of the population in R6, of both in R7). Mayor m approves if its gain over the reversion, a per-capita split of all the county money, is non-negative: $G_m = (1 + \ell)(t_m - R_m) + \sum_k \left[(1 + \tilde{\gamma}_m) w_{mk} V_k q_k - p_{mk} stake_{mk}\right]$, where t_m is its transfer and R_m its reversion. If the majority is not there, the conde compensates the cheapest majority, paying each member of the coalition its deficit out of the transfers of the others (Romer & Rosenthal, 1978; Baron & Ferejohn, 1989). If even that is infeasible, the lowest-ranked project is dropped and the procedure is repeated.

S2.4.8. Effort

The conde chooses measured supervision effort e on the grid $\{0, 0.1, \ldots, 1\}$ and unmeasured quality effort e_u in closed form (Proposition 5) to maximize

$$\theta\left[soc(e) + \alpha_u e_u Y(e)\right] + (1 - \theta)\, pol(e) + V_R \Pr(R \mid e) - c_e (e + e_u)^2 / 2, \quad \text{(S10)}$$

in units of the term's county money. Here soc and pol are the two braces of equation (S9) summed over the plan, Y(e) is the value of completed projects and $\Pr(R \mid e) = \Lambda\left(\text{logit}\, p_j^{next} + \rho_R \bar{g}(e)\right) \Pr(pass \mid e)$ is the probability of reappointment, with $\bar{g}(e)$ the mean gain of the college (by mayor, or by population in R6).

Without the bar, the pass probability is one; with a single term, $\Pr(R \mid e) = 0$; and for a technical conde, $\Pr(R \mid e) = \Pr(\text{pass} \mid e)$.

S2.4.9. Realization

Cooperation and completion are drawn. Completed projects deliver $(1 - d_c)V_k(1 + \alpha_u e_u)$, shared by w_{mk}. Unfinished projects are carried into the next term: they are abandoned with probability zero if the same conde continues, a_{same} if a co-partisan succeeds and a_{ab} if the party changes. A continued project is completed with the probability it had, and is then worth β_d times its value; an abandoned or failed one is worth v_{inc} per real spent. Projects unfinished at the end of 2038 are valued at their expected continuation value.

S2.4.10. Pre-election tilt

In the first half of the term, a conde who may be reappointed can move transfers from rival to co-partisan municipalities to improve the next college. The tilt solves Proposition 6, at a private cost θc_T per real moved relative to the term's money, and it is limited to half of the rivals' first-half transfers.

S2.4.11. Municipal elections

The statewide partisan climate at each election is a Dirichlet draw with mean $\bar{w}$ and precision c_{dir}. The mean requires an adjustment. Because a seat that changes hands goes to another party (as described below), a climate whose mean equals the 2024 party shares c would shrink the large parties and enlarge the small ones over time. The mean therefore solves $\bar{w}_p \sum_{q \neq p} c_q / (1 - \bar{w}_q) = c_p$ for every party p, which keeps the expected party composition at its 2024 level when every party's seats change hands at the same rate (PL's weight is 0.37, against a 2024 share of 0.30 in the climate's support, which reserves 1% for parties without mayors).

The elections then proceed seat by seat. An eligible incumbent runs with probability r_{run}. Before running, with probability h_{sw} the incumbent draws a new affiliation: the conde's party with probability β_{bw}, and otherwise a party drawn from the climate, which may be the incumbent's own, so that the probability of an actual change is below h_{sw} (h_{sw} is calibrated to the share of re-elected mayors who changed party; Supplementary Material 04, Section S4.5). A running incumbent wins with probability $\Lambda(\text{logit}\, p_{win} + \alpha_V\, g(x_m))$, where x_m is the change in the money the municipality received in the two pre-election years relative to the status quo, as a share of its revenue in those two years, and $g(x) = h_V \ln(1 + x/h_V)$ for $x \geq 0$ and $-h_V \ln(1 - x/h_V)$ for $x < 0$, which gives diminishing returns. An open seat is kept by the outgoing party with probability $\Lambda\left(\text{logit}\, r_{keep} + \alpha_V g(x_m)\right)$. Otherwise, and when an incumbent is defeated, the seat goes to another party: to the conde's party with probability β_{bw}, if it is not the outgoing party, and otherwise to a party drawn from the climate among the other parties. This rule follows the data, since in Santa Catarina none of the seats

whose incumbent ran and lost stayed with the incumbent's party in 2020 or 2024. Re-elected mayors become term-limited.

S2.4.12. Targets and sanctions

At the end of the term, the county's score is $z_j = \sqrt{w_c}\,\mathrm{cap}_j + \sqrt{w_p}\,\mathrm{perf}_j + \sqrt{w_n}\,\xi_j$, with weights normalized to sum to one ($w_p = \max\{0.1, 1 - w_c - w_n\}$ before normalization) and $\xi_j \sim N(0,1)$. In this score, cap_j is the standardized log GDP per capita of the county, or its deviation from the mean of its capacity tercile under peer comparison (R8), and perf_j is its standardized delivery, the share of the term's investment (regional projects and the investment part of local transfers) that was completed. The threshold is set so that a share ϕ_c of the counties would fail at baseline. A county that fails bars its conde from reappointment and, where the territorial cut applies, loses a share $s_T\Phi_j$ of its next-term money, that is, the cut times the probability Φ_j that the Union enforces it (Supplementary Proposition 8, Supplementary Material 03, Section S3.8). The money withheld is shared among the other counties.

S2.5. The status quo and the reference designs

Three reference designs isolate parts of the reform, and the status quo provides the common benchmark. In the status quo (SQ), the money follows the 2025 geography in every year, and municipal elections take place with the same random numbers, without counties, bandwagon or money effects. Closed lists only (CL) changes the Chamber and nothing else. The per-capita design (D0) abolishes amendments and transfers the county money to municipalities in proportion to population, without condes, projects or administration; its relocated share is the same as in the reform. Finally, L0 adds the managerial law to the status quo.

S2.6. The managerial law

Technical assistance raises the completion of local works by $g_{TA}(1 - \kappa_m)$ and of regional works by $g_{TA}(1 - \kappa_j)$, in proportion. Inspection multiplies diversion and capture by $I = (1 - p_I)[1 - p_I(1 + s_D)]^+$ (Supplementary Proposition 9, Supplementary Material 03, Section S3.9). The commissioners cost a share c_A of all the money they oversee, which is charged to municipalities in proportion to the value they receive.

S2.7. Value

For each world and design, the value received by municipality m is the sum over the twelve years of four components: its local transfers and relocated money, valued at v_m; its benefit shares of the value of regional projects net of capture, whether completed on time or a term late, together with the salvage and continuation values at the end; minus its co-financing, valued at v_m; and minus its share of the commissioners' cost. Real

value is the sum over municipalities. Need-weighted value weights municipality m by n_m^{ζ}, where n_m is need (the inverse of GDP per capita, normalized to a population-weighted mean of one). Both are reported as percentage changes relative to the status quo in the same world.

S2.8. Computation

The model is vectorized over worlds. Each world draws the 71 parameters from the priors of Table S4 (Supplementary Material 04) by a scrambled Sobol' sequence, together with a set of random numbers (votes, project costs and values, hosts, cooperation and completion shocks, election shocks, partisan climates and score noise) that is the same for every design (common random numbers). The 2,000 worlds are run in chunks of 500, and every design is run on every world.

Table S1. Notation of the model

Symbol	Meaning
M	Money, R$ 595.2 million a year (16 deputies' quotas)
x_m^{SQ}	Status-quo amendments to municipality m, R$ a year
η	Share of the abolished money re-created in discretionary channels
s_0	Status-quo share of deputies' effort on brokerage
π, κ	List-makers' weights on brokerage and on legislation, relative to voters'
t_0, ρ	Voters' weight on brokerage skill; correlation of legislative and brokerage skill
σ_ε, σ_ν	Noise of personal votes; weight of list-makers' own criteria
χ	Responsiveness of a co-partisan conde to deputies' requests
ψ_p, X_p	Surviving brokerage technology of party p; list-makers' relative weight on brokerage
$\bar{s}_p$	Share of county money controlled by party p's condes
θ, a_c	Conde's weight on value; competence
V_R	Value of reappointment to the conde (share of the term's money)
φ_P, ρ_R	Party loyalty and retrospective weight in the college vote
$\bar{s}$, $\bar{\sigma}$, sd_σ	Cost of the menu relative to county money; mean and dispersion of net scale economies
w_{mk}, $\tilde{w}_{mk}$	Benefit share; co-financing share among participants
χ_{cof}	Co-financing share (0.3)
ℓ	Mayors' localism premium
ϵ_x	Share of a non-payer's benefits that can be withheld
$\tilde{\gamma}_m$, ζ_{opp}	Mayor's credit for regional projects; discount for mayors of other parties
ς	Scale of private cost shocks relative to the stake
δ_m, π_m^{stay}, π_j^{cont}	Mayor's discount on future projects; probability of staying in office; continuity of the conde's office
κ_P	Share of commitments honored by a co-partisan successor
$\bar{q}$, ν, ϕ_{cx}	Completion of transfer-funded works; elasticity to capacity; penalty of regional works
κ_m, κ_j, g_{pool}	Capacity of a municipality and of a county; capacity gap closed by a pooled team
α_e, α_a, α_u	Effects of supervision effort, competence and unmeasured quality effort
c_e	Cost of effort

Symbol	Meaning
ω_h, ν_c	Completion lost if the host obstructs; elasticity of completion to the funded share
v_{inc}, a_{ab}, a_{same}, β_d	Salvage value; abandonment after a party change or within the party; value of late completion
d_m, d_c	Municipal diversion; county capture
d_{c0}, D_{max}, g_{ten}	Baseline capture; discipline of a contestable reappointment; rise with tenure
r_{run}, p_{win}, h_{sw}, r_{keep}	Incumbents who run; runners who win; switching hazard; open seats kept
β_{bw}, c_{dir}	Bandwagon toward the conde's party; precision of the statewide climate
α_V, h_V	Electoral return to local money; scale of diminishing returns
c_T	Conde's cost per real moved before the election
w_c, w_n, ϕ_c	Weights of capacity and noise in the county score; baseline failure rate
s_T, Ξ, ϱ	Share of money withheld; Union's resolve; need gradient of the harm
p_I, s_D, g_{TA}, c_A	Inspection efficacy; penalty; technical assistance; commissioners' cost
n_m, ζ	Need; weight on need

Supplementary Material 03 – Proofs and Numerical Checks

This unit proves the seven propositions of Section 4 of the article (Sections S3.1–S3.7) and two supplementary propositions that the model uses, on the credibility of territorial sanctions and on inspection (Sections S3.8 and S3.9). Sections S3.1–S3.7 follow the same order, so that the reader can find each part in the same place: each first restates the proposition, then gives the idea of the proof in a few sentences, and finally proves the proposition in numbered steps, with the main expressions displayed and numbered; where it helps, a remark connects the result to the article. Sections S3.8 and S3.9, whose proofs take a few lines, give only the statement and the proof. Because closed forms can hide errors of algebra or of implementation, each of Propositions 1–7 was also checked by brute force against the engine or against direct numerical optimization. Section S3.10 and Table S2 report these checks; their code is in code/cc2_props.py and their results in out/cc2_props.json of the replication package.

S3.1. Proposition 1 (selection)

S3.1.1. Statement

Suppose that a party elects the S highest-ranked of N aspirants on the index $I = \lambda_L + t\lambda_B + s\epsilon$, where (λ_L, λ_B) is standard bivariate normal with correlation ρ and ϵ is an independent standard normal. Then the expected legislative skill of the elected is

$$\mathbb{E}[\lambda_L \mid \text{elected}] = r(t, s)\, a(S, N), \tag{S11}$$

where $a(S, N)$ is the expected mean of the S largest of N standard normal draws and

$$r(t, s) = \frac{1 + t\rho}{\sqrt{1 + t^2 + 2t\rho + s^2}}. \tag{S12}$$

The function r falls with t if and only if $t > \rho s^2/(1 - \rho^2)$, and closed lists raise the expected legislative skill of the elected if and only if $r(Xt_0, \sigma_\nu) > r(t_0, \sigma_\varepsilon)$.

S3.1.2. Idea of the proof

Selection depends only on the index, so the skill of the elected depends on how closely the index tracks skill. For normal variables, the expected skill given the index is proportional to the standardized index, with the correlation between the two as the factor of proportionality, while the expected mean of the selected standardized indices does not depend on the weights in the index.

S3.1.3. Proof

Step 1 (one aspirant)

The index has variance $D = 1 + t^2 + 2t\rho + s^2$ and covariance $1 + t\rho$ with λ_L. Because (λ_L, I) is bivariate normal,

$$\mathbb{E}[\lambda_L \mid I] = r(t, s)\,\tilde{I}, \tag{S13}$$

where $\tilde{I} = I/\sqrt{D}$ is the standardized index and $r(t, s) = (1 + t\rho)/\sqrt{D}$ is the correlation between legislative skill and the index.

Step 2 (the elected)

Aspirants are independent, so, conditional on the vector of indices, the legislative skill of each aspirant has mean $r\tilde{I}_i$ and does not depend on the other aspirants' indices. The selection event depends only on the indices. Hence

$$\mathbb{E}[\lambda_L \mid \text{elected}] = r\,\mathbb{E}[\text{mean of the S largest } \tilde{I}_i] = r\,a(S, N), \tag{S14}$$

since the $\tilde{I}_i$ are independent standard normal. The same argument with $\mathrm{Cov}(\lambda_B, I) = t + \rho$ gives the expected brokerage skill of the elected, $\mathbb{E}[\lambda_B \mid \text{elected}] = r_B\,a(S, N)$, with $r_B = (t + \rho)/\sqrt{D}$.

Step 3 (the weight on brokerage)

Differentiating r with respect to t,

$$\frac{\partial r}{\partial t} = D^{-3/2}[\rho D - (1 + t\rho)(t + \rho)] = D^{-3/2}[\rho s^2 - t(1 - \rho^2)], \tag{S15}$$

which is negative if and only if $t > \rho s^2/(1 - \rho^2)$.

Step 4 (closed against open lists)

Open lists rank aspirants with $t = t_0$ and $s = \sigma_\varepsilon$, and closed lists with $t = Xt_0$ and $s = \sigma_\nu$, while $a(S, N)$ is the same under both rules. Closed lists therefore raise the expected legislative skill of the elected if and only if $r(Xt_0, \sigma_\nu) > r(t_0, \sigma_\varepsilon)$. ▫

S3.1.4. Remark

A lower weight on brokerage acts through the numerator and the denominator of r jointly, so it does not guarantee better legislators. If list-makers' own criteria are noisier than personal votes ($\sigma_\nu > \sigma_\varepsilon$), $r(Xt_0, \sigma_\nu)$ can fall below $r(t_0, \sigma_\varepsilon)$ for every X, which is the case of the lowest curve of Figure 3a of the article.

S3.2. Proposition 2 (effort and relocation)

S3.2.1. Statement

After the reform, a deputy of party p devotes a share

$$e_{B,p} = \frac{X_p s_0}{X_p s_0 + 1 - s_0} \tag{S16}$$

of effort to brokerage, where $X_p = \pi\psi_p/\kappa$ and $\psi_p = \eta + \chi(1-\eta)\bar{s}_p$ is the brokerage technology that survives the abolition of amendments (equation (S3) in Supplementary Material 02). Legislative effort rises if and only if $X_p < 1$. If the whole abolished amount reappears in discretionary channels ($\eta = 1$), abolition adds nothing to closed lists; otherwise, and if co-partisan condes respond to deputies ($\chi > 0$), parties whose condes control more county money keep more brokerage.

S3.2.2. Idea of the proof

With logarithmic payoffs, the share of effort devoted to brokerage is the weight on brokerage divided by the sum of the two weights. The reform multiplies the weight on brokerage by $\pi\psi_p$ and the weight on legislation by κ, and the status-quo share s_0 pins down the ratio of the original weights, so the new share depends only on s_0 and on the ratio X_p of the two multipliers.

S3.2.3. Proof

Step 1 (the status quo)

The objective $W_B \ln e + W_L \ln(1-e)$ is strictly concave on $(0,1)$, and the first-order condition gives $e = W_B/(W_B + W_L)$. In the status quo this share is s_0, so $W_L = W_B(1-s_0)/s_0$.

Step 2 (after the reform)

After the reform the weights are $\pi\psi_p W_B$ and κW_L, so that, substituting W_L from Step 1,

$$e_{B,p} = \frac{\pi\psi_p W_B}{\pi\psi_p W_B + \kappa W_L} = \frac{X_p s_0}{X_p s_0 + 1 - s_0}, \tag{S17}$$

with

$$X_p = \frac{\pi\psi_p}{\kappa}. \tag{S18}$$

Step 3 (the condition)

The share $e_{B,p}$ is increasing in X_p and equals s_0 at $X_p = 1$. Legislative effort $1 - e_{B,p}$ therefore exceeds $1 - s_0$ if and only if $X_p < 1$.

Step 4 (relocation and the condes)

If $\eta = 1$, then $\psi_p = 1$ for every party and $X_p = \pi/\kappa$, the value under closed lists without abolition. For $\eta < 1$ and $\chi > 0$, $\psi_p = \eta + \chi(1-\eta)\bar{s}_p$ is increasing in $\bar{s}_p$, so parties whose condes control more county money keep more brokerage. ▫

S3.3. Proposition 3 (cooperation)

S3.3.1. Statement

Given the plan and the other mayors' choices, mayor m co-finances project k (or, as its host, provides land and permits) with probability $\Lambda\big(NG_{mk}/(\varsigma\, stake_{mk})\big)$, where Λ is the logistic function, ς the scale of the cost shock and

$$NG_{mk} = (1+\tilde{\gamma}_m)\, w_{mk} V_k\, (\Delta q_{mk} + \epsilon_x\, q_{mk}) + \delta_m F_m - stake_{mk} \tag{S19}$$

is the mayor's net gain from cooperating. Cooperation is more likely than not if and only if $\delta_m > \delta^*_{mk}$, with

$$\delta^*_{mk} = \frac{stake_{mk} - (1+\tilde{\gamma}_m)\, w_{mk} V_k\, (\Delta q_{mk} + \epsilon_x\, q_{mk})}{F_m}. \tag{S20}$$

S3.3.2. Idea of the proof

The mayor compares the payoff of cooperating with the payoff of not cooperating. The difference has three parts, namely what happens to the current project, what happens to the mayor's future participation and what cooperation costs, and a logistic cost shock turns this difference into a probability.

S3.3.3. Proof

Step 1 (the net gain)

Fix the strategies of the other mayors and the plan. Cooperation changes the mayor's payoff in three ways. First, it raises the completion probability of the project by Δq_{mk}, which raises the mayor's expected benefit by $(1+\tilde{\gamma}_m) w_{mk} V_k \Delta q_{mk}$; in addition, a mayor who does not pay loses a share ϵ_x of the municipality's benefits from the project if the project is completed. The model values this loss at q_{mk}, the completion probability when m cooperates; valued at the completion probability without m's cooperation, $q_{mk} - \Delta q_{mk}$, it

would be smaller by $\epsilon_x(1+\tilde{\gamma}_m)w_{mk}V_k\Delta q_{mk}$, a term of second order in the share of a small participant (see the corollary below). Equation (S19) is therefore the net gain as the model defines it, with this approximation. Second, a mayor who obstructs is excluded from the next term's projects, worth F_m, if the mayor is still in office and the conde's office keeps its commitments, which happens with probability $\pi_m^{stay}\pi_j^{cont}$; discounted by β_y^4 over the four years of a term, this loss is $\delta_m F_m$, with $\delta_m = \beta_y^4\pi_m^{stay}\pi_j^{cont}$. Third, cooperation costs the co-financing share, valued at $1+\ell$ per real, plus, for the host, the local cost; together these make up stake_{mk}. Adding the three parts gives equation (S19).

Step 2 (the probability)

The mayor cooperates if $NG_{mk} + \epsilon_{mk} \geq 0$, where ϵ_{mk} is the private cost shock, logistic with scale $\varsigma\,\text{stake}_{mk}$. Hence the probability of cooperation is $\Lambda\big(NG_{mk}/(\varsigma\,\text{stake}_{mk})\big)$, which exceeds one half if and only if $NG_{mk} > 0$. Solving $NG_{mk} > 0$ for δ_m, with $F_m > 0$, gives $\delta_m > \delta_{mk}^*$, with the threshold of equation (S20). ▫

Existence of an equilibrium

The map from the vector of cooperation probabilities to the vector of best-response probabilities is continuous from the cube $[0,1]^n$ into itself, so a fixed point exists (Brouwer). The engine iterates best responses six times from $p = 0.7$, which leaves changes of the order of 10^{-4} in the last iteration; iterated 60 times, the residual falls below 10^{-15} (Table S2).

S3.3.4. Corollary (small participants)

For a mayor with a small share of a project, the benefit that the mayor's own payment buys through completion is second order in the share, while the payment is first order. The sign of NG_{mk} is then decided by the enforcement terms against the stake.

Proof of the corollary

Let $c = (1-\chi_{cof}) + \chi_{cof}A_{-m}$ be the funded share without m's payment. Then

$$\Delta q_{mk} = Q\,H_k[(c+\chi_{cof}\tilde{w}_{mk})^{\nu_c} - c^{\nu_c}] = Q\,H_k\,\nu_c\,c^{\nu_c-1}\chi_{cof}\tilde{w}_{mk} + O\big(\tilde{w}_{mk}^2\big) \qquad \text{(S21)}$$

is first order in the mayor's share, so the benefit of completion, $w_{mk}V_k\Delta q_{mk}$, is second order in it, while the stake, $(1+\ell)\chi_{cof}C_k\tilde{w}_{mk}$, is first order. For small shares, the sign of NG_{mk} is therefore decided by the enforcement terms, $\epsilon_x w_{mk}V_k q_{mk}$ (first order) and $\delta_m F_m$, against the stake. ▫

S3.3.5. Comparative statics

NG_{mk} is increasing in ϵ_x, δ_m, F_m, $\tilde{\gamma}_m$ and V_k, and decreasing in the stake and in ℓ. The horizon δ_m is zero for term-limited mayors. The continuity term $\pi_j^{cont} = p_j^{next}[f + \kappa_P(1 - f)]$ is lower with a single term ($f = 0$) than with the bar ($f = f_j$, the probability that the county meets its targets), and lower with the bar than without it ($f = 1$), when $\kappa_P < 1$; with the bar, it is lower in counties less likely to meet their targets. These comparative statics are the basis of the results on cooperation in Section 6.5 of the article.

S3.4. Proposition 4 (completion hurdle)

S3.4.1. Statement

A conde who values the money at face value ($\theta = 1$) chooses project k, considered on its own, if and only if its net economies of scale exceed a hurdle, $\sigma_k > \sigma^*(\tilde{q}_k)$, with

$$\sigma^*(\tilde{q}_k) = \frac{f_k\left[\bar{v}_j/(1 - d_c) - s_w(\tilde{q}_k)\right]}{q_{eff}(\tilde{q}_k)} - 1, \tag{S22}$$

where $\bar{v}_j$ is the value per real of local money in the county, d_c the diversion and capture in county procurement, f_k the expected funded share of the project, q_{eff} the probability of completion now or, with a delay, next term, and s_w the expected salvage value of unfinished works, the last two defined in the proof. The hurdle falls with expected completion unless the salvage value of unfinished works is large.

S3.4.2. Idea of the proof

A real spent on the project competes with a real transferred to the municipalities. The project pays its full value if it is completed on time and a discounted value if it is completed a term late, and otherwise yields only a salvage value; comparing the two uses of the money and solving for the scale economies gives the hurdle.

S3.4.3. Proof

Step 1 (the inclusion rule)

Consider a project in isolation, with budget available and the plan approved. A conde with $\theta = 1$ includes project k if and only if its social value exceeds the value of the local money it displaces:

$$(1 - d_c)\left[qV + (1 - q)\,U^{unf}\right] > \bar{v}\,fC, \tag{S23}$$

where fC is the money spent on the project (f the expected funded share of its cost C), q its completion probability and

$$U^{unf} = (1 - \bar{a})(q\beta_d V + (1 - q)\, v_{inc} fC) + \bar{a}\, v_{inc} fC \tag{S24}$$

the expected value of an unfinished project: it is continued with probability $1 - \bar{a}$ and then completed with probability q, at a value discounted by β_d, and it is otherwise worth its salvage value v_{inc} per real spent.

Step 2 (collecting terms)

Writing $V = (1 + \sigma)C$ and collecting terms, the inclusion rule becomes

$$(1 - d_c)\, C\, [(1 + \sigma)\, q_{eff} + f\, s_w] > \bar{v}\, fC, \tag{S25}$$

where

$$q_{eff} = q + (1 - q)(1 - \bar{a})\, q\, \beta_d \tag{S26}$$

is the probability of completion on time or, at a discount, a term late, and

$$s_w = (1 - q)[(1 - \bar{a})(1 - q) + \bar{a}]v_{inc} \tag{S27}$$

is the expected salvage value of unfinished works per real spent.

Step 3 (the hurdle)

Dividing both sides of equation (S25) by $(1 - d_c)\, C$ and solving for σ shows that the rule holds if and only if $\sigma > f[\bar{v}/(1 - d_c) - s_w]/q_{eff} - 1$, the hurdle of equation (S22).

Step 4 (how the hurdle varies with completion)

The effective completion increases with q, since

$$\frac{\partial q_{eff}}{\partial q} = 1 + (1 - \bar{a})\, \beta_d\, (1 - 2q) \tag{S28}$$

is positive for every q in $[0,1]$, because $(1 - \bar{a})\beta_d < 1$. The salvage term decreases with q, since

$$\frac{\partial s_w}{\partial q} = -v_{inc}[2(1 - \bar{a})(1 - q) + \bar{a}] \tag{S29}$$

is negative. The two effects work in opposite directions on the hurdle. Differentiating the hurdle with respect to q shows that it falls with q if and only if

$$v_{inc}[2(1-\bar{a})(1-q)+\bar{a}]\,q_{eff} < [\bar{v}/(1-d_c)-s_w]\,\frac{\partial q_{eff}}{\partial q}, \tag{S30}$$

that is, unless the salvage value of unfinished works is large relative to the value of a real of local money. ▫

S3.4.4. Remark

The condition of equation (S30) held in all of 5,000 parameter draws from the priors, evaluated at $q = 0.4$ (Table S2), so at that completion rate the hurdle falls with completion in every draw. The same property underlies the result of Section 6.4 of the article that, at the completion rates of Brazilian transfer-funded works, a regional project must replace a great deal of duplicated spending to beat local spending.

S3.5. Proposition 5 (effort and crowding out)

S3.5.1. Statement

(i) Unmeasured effort is

$$e_u^* = \max\{0,\ \theta\alpha_u Y/c_e - e_m\}, \tag{S31}$$

where Y is the value of completed projects per unit of the term's money, so every unit of measured effort induced by reappointment or targets displaces one unit of unmeasured effort until none is left. (ii) With the proposal's bar, the reappointment motive adds

$$V_R \Pr(\text{win})\,\varphi\big((\mu(e_m)-\bar{z})/\sigma_z\big)\,\mu'(e_m)/\sigma_z \tag{S32}$$

to the marginal return of measured effort, where $\mu(e_m)$ is the county's expected score, $\bar{z}$ the threshold and σ_z the noise of the score; the motive is strongest for counties whose expected score is at the threshold and vanishes for those far above or below it. (iii) With one vote per mayor, the probability of winning depends on the average gain of the municipalities, each counting equally whatever its population.

S3.5.2. Idea of the proof

The two efforts are substitutes in the conde's cost, so the first-order condition for unmeasured effort fixes their sum. The bar adds a motive that acts through the probability of passing the target, whose slope is proportional to the normal density at $(\mu - \bar{z})/\sigma_z$.

S3.5.3. Proof

Step 1 (crowding out)

Unmeasured effort enters the conde's objective as $\theta\alpha_u e_u Y - c_e(e_m + e_u)^2/2$. The objective is strictly concave in e_u, and the first-order condition

$$\theta\alpha_u Y = c_e(e_m + e_u) \tag{S33}$$

gives equation (S31); for $e_u^* > 0$ and holding Y fixed, $\partial e_u^* / \partial e_m = -1$.

Step 2 (where the motive acts)

With the bar, the probability of reappointment is

$$\Pr(R \mid e_m) = \Pr(\text{win})\, \Phi\big((\mu(e_m) - \bar{z})/\sigma_z\big), \tag{S34}$$

whose derivative with respect to e_m, taking Pr(win) as given and multiplied by V_R, is expression (S32), the contribution of the reappointment motive to the marginal return of measured effort. The normal density is maximal at zero and vanishes in both tails, so the motive is largest when the expected score is at the threshold.

Step 3 (whose gains count)

With one vote per mayor, Pr(win) depends on the mean of the mayors' gains, each with weight $1/n_j$ whatever the municipality's population. ▫

S3.6. Proposition 6 (pre-election tilt)

S3.6.1. Statement

With electoral returns that diminish in the money received, the conde's tilt toward co-partisan municipality m is

$$T_m = (A_m/\lambda - h_V \, \mathrm{Rev}_m)^+, \tag{S35}$$

where A_m is proportional to the value of reappointment, the probability that one more co-partisan mayor changes the next college's plurality, the uncertainty of the local race and the electoral return to money, λ is the conde's cost per real moved and Rev_m the municipality's revenue in the two pre-election years. At the margin, raising the expected number of co-partisan mayors by one costs $Rev_m/[p_m(1-p_m)\alpha_V]$.

S3.6.2. Idea of the proof

Each real moved to a co-partisan municipality raises the chance that its next mayor belongs to the conde's party, with diminishing returns, and costs the conde a constant amount. The optimal tilt equates the two at the margin, municipality by municipality, and is zero where even the first real moved does not pay.

S3.6.3. Proof

Step 1 (the electoral gain)

Let $T_m \geq 0$ be the money moved to co-partisan municipality m in the two pre-election years. The probability that the conde's party keeps or wins the municipality is

$$p_m(T_m) = \Lambda\big(a_m + \alpha_V\, g(T_m/Rev_m)\big), \tag{S36}$$

with $g(x) = h_V \ln(1 + x/h_V)$, which gives diminishing returns.

Step 2 (the conde's problem)

Linearizing Λ around the status quo, the expected gain in the value of reappointment is

$$\sum_m V_R\ piv\, \omega_m\, p_m(1-p_m)\, \alpha_V\, h_V \ln\big(1 + T_m/(h_V Rev_m)\big), \tag{S37}$$

where piv is the probability that one more co-partisan mayor changes the plurality of the next college and ω_m the mayor's vote weight. With a cost λ per real moved, the conde maximizes this gain minus $\lambda \sum_m T_m$, which is a concave problem.

Step 3 (the solution)

The Kuhn–Tucker conditions give $A_m/(h_V Rev_m + T_m) = \lambda$ if $T_m > 0$, with $A_m = V_R\, piv\, \omega_m\, p_m(1-p_m)\, \alpha_V\, h_V$. Hence $T_m = (A_m/\lambda - h_V Rev_m)^+$, which is equation (S35), the water-filling solution.

Step 4 (the price of control)

At $T_m = 0$, the derivative of p_m with respect to T_m is $p_m(1 - p_m)\alpha_V/Rev_m$, because $g'(0) = 1$. Raising p_m by one unit therefore costs $Rev_m/[p_m(1 - p_m)\alpha_V]$, which is the price of one more expected co-partisan mayor. ▫

S3.6.4. Implementation

When the total tilt would exceed half of the rival municipalities' first-half transfers, the engine scales it down proportionally.

S3.6.5. Magnitude

For the median municipality of Santa Catarina, with current revenue of R$ 54.7 million a year and an even race, the price of one more expected co-partisan mayor is R$ 135 million at the midpoint of the prior of α_V and R$ 73 million at its upper bound. These are the magnitudes that Section 6.2 of the article compares with the money of a county.

S3.7. Proposition 7 (the sanction trap)

S3.7.1. Statement

Consider a county in which completion c determines the probability of meeting the target, $\Pi(c)$, increasing in c, which determines the continuity of the conde's office, which in turn determines cooperation and therefore next-term completion:

$$c' = \Gamma(c) = Q\,\Lambda\big((g_0 + \beta\pi_w[\kappa_P + (1 - \kappa_P)\Pi(c)]F)/\varsigma\big), \qquad \text{(S38)}$$

where $Q < 1$ is technical completion, g_0 the net gain of cooperation without enforcement, β the discount factor of a term, π_w the probability that the conde's party keeps the office, F the value of future projects, ς the scale of the cost shock and the bracket the continuity of the conde's office when a conde who misses the target is barred. (i) Without the bar, Γ does not depend on c and the steady state is unique. (ii) With the bar and $\kappa_P < 1$, Γ lies below its no-bar value for every c. (iii) When Γ is steep enough to cross the diagonal from below ($\Gamma' > 1$ at the crossing), there are two stable steady states, and a county with low capacity can be locked into low completion and repeated failure; peer comparison moves the threshold toward each county's own capacity.

S3.7.2. Idea of the proof

Without the bar, the continuity of the conde's office does not depend on performance, so the completion map is flat. With the bar, low completion lowers continuity, lower continuity weakens cooperation and weaker cooperation lowers completion, so the map slopes upward; if it crosses the diagonal from below, it must cross it three times.

S3.7.3. Proof

Step 1 (no bar)

Without the bar, the bracket in equation (S38) is one, so Γ is constant and the steady state is unique.

Step 2 (the bar lowers the map)

With the bar and $\kappa_P < 1$, the bracket $\kappa_P + (1 - \kappa_P)\Pi(c)$ is below one whenever $\Pi(c) < 1$, which holds for every c because Π is a normal distribution function (Step 4). Since Λ is increasing, Γ lies below its no-bar value.

Step 3 (steady states)

Γ is continuous and increasing, with $\Gamma(0) > 0$ and $\Gamma(1) < 1$; hence it crosses the diagonal at least once, and every crossing at which $\Gamma' < 1$ is a stable steady state. If Γ crosses the diagonal from below at some $\hat{c}$, where $\Gamma'(\hat{c}) > 1$, there must be a crossing from above on each side of $\hat{c}$, so there are two stable steady states separated by an unstable one.

Step 4 (capacity and peer comparison)

The probability of meeting the target is

$$\Pi(c) = \Phi\big((\mu + bc - \bar{z})/\sigma\big), \tag{S39}$$

which depends on the county's capacity through μ. Low-capacity counties have Π close to zero at low completion and can be held at the low steady state, with low completion and repeated failure. Comparing counties with peers of similar capacity raises $\mu - \bar{z}$ for low-capacity counties and can remove the low steady state. ▫

S3.7.4. Example

With $Q = 0.9$, $g_0 = -1.5$, $\beta = 0.6$, $\pi_w = 0.9$, $F = 4$, $\varsigma = 0.4$, $b = 6$, $\sigma = 0.5$, $\bar{z} = 0$ and $\kappa_P = 0.2$ (Figure 8b of the article), there are three steady states for capacity $\mu \in [-3.5, -1.15]$. At $\mu = -2.5$, for example, they are 0.058, 0.439 and 0.755, while without the bar the only steady state is 0.755.

S3.8. Supplementary Proposition 8 (credibility of territorial sanctions)

S3.8.1. Statement

If the Union's private benefit from enforcing a cut on a failing county is uniform on $[0, \Xi]$, and the political cost of enforcing it, borne because the harm falls on the county's population, is n_j^{ϱ}, where n_j is the county's need, then the cut is enforced with probability

$$\Phi_j = \left(1 - n_j^{\varrho}/\Xi\right)^+, \tag{S40}$$

which falls with need when $\varrho > 0$.

S3.8.2. Proof

The Union enforces the cut if and only if its benefit exceeds the cost, which happens with probability $\Pr\left(U[0, \Xi] > n_j^{\varrho}\right) = \left(1 - n_j^{\varrho}/\Xi\right)^+$. ▫

S3.8.3. Remark

The proposition formalizes the soft budget constraint (Kornai et al., 2003): sanctions that harm the population are least credible where the population is neediest.

S3.9. Supplementary Proposition 9 (inspection)

S3.9.1. Statement

Suppose that each real of potential diversion is attempted if the expected return exceeds a private cost $c \sim U[0,1]$, that inspection detects and blocks a share p_I of attempts, and that a detected attempt pays a penalty s_D per real. Then diversion is multiplied by

$$I = (1 - p_I)[1 - p_I(1 + s_D)]^+. \tag{S41}$$

S3.9.2. Proof

An attempt yields one real with probability $1 - p_I$ and costs s_D with probability p_I, so its expected return is $1 - p_I(1 + s_D)$, and the share of potential diversion attempted is $\Pr\big(c < 1 - p_I(1 + s_D)\big) = [1 - p_I(1 + s_D)]^+$ (Becker, 1968). Of the attempts, a share $1 - p_I$ succeeds, which gives equation (S41). Without inspection ($p_I = 0$), the factor is one. ▫

S3.10. Numerical checks

Table S2 reports the brute-force checks. For each proposition, the closed form or the rule stated in the proposition was compared with a direct computation. Proposition 1 was checked by simulating the selection of aspirants, Propositions 2 and 5 by a grid search over effort, and Proposition 3 by simulated cost shocks and by the equilibrium that the engine computes. Proposition 4 was checked by a direct evaluation of the inclusion rule over random draws, Proposition 6 by random perturbations of the tilt and Proposition 7 by a numerical search for the steady states of the completion map. No check found a violation of a stated rule, and the errors of the closed forms are of the order of the grid step (Propositions 2 and 5) or of the simulation noise (Proposition 1).

Table S2. Brute-force checks of the propositions

Check	Result
P1: maximum absolute error of the closed form for legislative skill (40 random cases, 60,000 replications each)	0.009
P1: maximum absolute error for brokerage skill	0.007
P1: sign violations of $\partial r/\partial t$ (2,000 draws)	0
P2: maximum absolute error of the closed-form effort (2,000 draws, grid search)	2.5×10^{-6}
P2: violations of "legislative effort rises iff $X < 1$"	0
P3: violations of the logistic best response and of the threshold (20,000 draws)	0
P3: residual of the cooperation fixed point after 60 iterations (the engine uses six)	3.9×10^{-16}
P3: mean change in cooperation when future value doubles / when excludability rises by 0.3	+0.078 / +0.063
P4: violations of the hurdle rule (20,000 draws)	0
P4: share of draws in which the hurdle falls with completion (5,000 draws)	1.00
P5: maximum absolute error of the closed-form unmeasured effort (3,000 draws)	5.0×10^{-6}
P5: location of the peak of the marginal pass incentive (threshold at 0)	0.00
P6: improvements on the water-filling solution found by random perturbation (300 problems)	0
P7: capacity range with three steady states, with the bar / without the bar	$[-3.5, -1.15]$ / none

Supplementary Material 04 – Data and Calibration

The model of the article combines quantities observed in Santa Catarina, parameters calibrated to the current regime and priors for institutions that do not exist yet. This unit documents the first two groups and lists all the parameters. Sections S4.1 to S4.3 describe the data: their sources, the construction of the variables and of the status quo, and what the data show. Sections S4.4 to S4.9 describe the calibration: the full list of parameters, the municipal electoral module, completion and the capacity gradient, diversion and capture, the electoral return to money and relocation.

S4.1. Sources

Table S3 lists the data, what each is used for and where it comes from. The mayoral, fiscal and amendment files were assembled for the companion evaluation of the managerial law (Companion paper, 2026) from copies of official data published in public repositories (the Superior Electoral Court, TSE; the National Treasury's SICONFI; the federal transfers platform Transferegov, through the repository of the NGO Transparência Brasil; and IBGE), and they were checked there against official totals. The votes for federal deputy come from the TSE's open-data repository: the files of votes by candidate, municipality and electoral zone of the 2022 and 2026 general elections in Santa Catarina (Tribunal Superior Eleitoral, 2026), both generated on 8 October 2026. The 2026 results are therefore those available four days after the election, and later judicial decisions on individual candidacies could change them.

Table S3. Data, their use in the model and their sources

Data	Use in the model	Source
Population of the 295 municipalities, 2022 Census	Per-capita split, weights, capacity index	IBGE, Censo Demográfico 2022
GDP per capita, 2021	Need; capacity term of the county score	IBGE, PIB dos Municípios
Municipal coordinates and boundaries	Distances between municipalities; maps	IBGE municipal mesh
17 health regions and 8 macro-regions	Counties (base design and robustness)	Comissão Intergestores Bipartite de Santa Catarina (2024), Deliberation 38/CIB/2024
Mayors elected in 2016, 2020 and 2024: party, re-election, candidacy, term limits	Real 2024 colleges; calibration of the municipal module	TSE results, via the companion package
Votes for federal deputy by candidate and municipality, 2022 and 2026; seats won in 2026	Status-quo geography of the three deputies who used special transfers little; Chamber module (seats); electoral connection (Supplementary Material 07, Section S7.2); status quo of the Chamber elected in 2026 (robustness)	TSE, open-data repository (votes by candidate, municipality and zone), summed over zones
Municipal current revenue and own-tax revenue, 2024	Electoral return to money; capacity index	SICONFI (National Treasury), via the companion package; imputed for 41 municipalities

Data	Use in the model	Source
Share of adults with higher education	Capacity index	IBGE, Censo Demográfico 2022, via the companion package
Revenue-sharing fund (FPM), 2024	Imputation of revenue	National Treasury, via the companion package
Special transfers ("emendas Pix") planned in 2025, by author and municipality, with technical impediments	Status-quo allocation; elasticity of completion to capacity	Transferegov, via Transparência Brasil and the companion package

S4.2. Construction of the variables and of the status quo

S4.2.1. Need

Need is the inverse of GDP per capita, normalized to a population-weighted mean of one.

S4.2.2. Capacity

The technical capacity of municipality m is $\kappa_m = 1/(1 + e^{-z_m})$, where z_m is the standardized mean of the standardized logarithm of population, the share of adults with higher education and the share of own taxes in current revenue. The logistic transformation keeps the index between zero and one. The mean index is 0.31 among municipalities with fewer than 5,000 inhabitants and 0.91 among those with more than 100,000.

S4.2.3. Revenue

Current revenue and the own-tax share are missing in SICONFI for 41 municipalities. They are imputed by ordinary least squares of the logarithm of revenue per capita (and of the own-tax share) on the logarithms of population, GDP per capita and FPM per capita and on the higher-education share, fitted on the 254 municipalities that report ($R^2 = 0.81$ for revenue).

S4.2.4. The status-quo allocation

For each deputy d, the share of the 2025 special-transfer plans that goes to municipality m, π_{dm}, is applied to the deputy's quota of R$ 37.2 million: $x_{dm}^{SQ} = 37.2\,\pi_{dm}$ million a year. Three deputies used special transfers little or not at all (R$ 2.8 million and R$ 5.9 million, and none). They are given the geography that their own 2022 votes predict under the targeting regression of Supplementary Material 07 (Section S7.2), fitted on the other thirteen. The predicted transfers of each of the three to each municipality, which depend on the deputy's share of the municipality's 2022 vote, its population, its GDP per capita and whether its mayor belongs to the deputy's party, are kept in the 29 municipalities with the largest predictions, the number funded by the median deputy, and normalized to the quota. In sample, the 29 municipalities with the largest predicted transfers

received a median of 43% of each of the thirteen deputies' actual money, and the median correlation between predicted and actual shares is 0.28. The imputed geography therefore describes where deputies with a given electoral base send money, and not each deputy's choices.

Four alternatives are used in the robustness checks of Section 6.10 of the article. The first gives the three small users the pooled geography of the thirteen. The second gives the pooled geography to every deputy. The third distributes half of each quota, the health half, by population and the other half with the deputy's own geography. The fourth uses the geography of the deputies elected in 2026: the nine re-elected deputies keep theirs, and the seven newcomers are given the geography predicted by their 2026 votes in the same way, with partners in a federation counted as co-partisans of its largest party. These alternatives change the status-quo distribution of amendments in the model. Amendments per inhabitant in the municipalities with fewer than 5,000 and more than 100,000 inhabitants are R$ 255 and R$ 32 in the main specification, R$ 284 and R$ 25 with the pooled geography for the three small users, R$ 299 and R$ 20 with the pooled geography for every deputy, R$ 167 and R$ 55 with half of each quota by population and R$ 155 and R$ 35 with the deputies elected in 2026. In the same order, the population-weighted Gini coefficients of amendments per inhabitant are 0.64, 0.72, 0.72, 0.32 and 0.57, and the neediest fifth of the population receives 2.0, 2.2, 2.2, 1.5 and 2.4 times its population share of the money.

Two further details complete the status quo. Two authors of special transfers had no party in the source and were matched by hand: Carol De Toni (PL, deputy) and Ivete da Silveira (MDB, senator). The 15 deputies who used special transfers planned R$ 243 million to the state's municipalities in 2025, and the three senators R$ 83 million.

S4.2.5. Seats

In 2026, PL won 7 of the state's 16 seats in the Chamber; the federation of PT, PCdoB and PV won 3 (two PT deputies and one PCdoB deputy); and MDB, Novo, PSD, Podemos, Republicanos and the federation of União and PP (with a PP deputy) won one each. A federation would present a single closed list, so the Chamber module gives its seats to its largest party in the state, PT and PP, whose condes define the county money that the federation's deputies can reach. In 2022 the seats were PL 6, MDB 3, PT 2, PSD 2 and Cidadania, Novo and União 1 each.

S4.2.6. Corrections

In the companion mayor files, the electoral codes of Santa Terezinha do Progresso and Santiago do Sul were matched to each other's statistical codes; the match was corrected, and the same crosswalk is used for the TSE vote files. Party labels follow the TSE at each election, and mergers (DEM and PSL into União, PTB and Patriota into PRD, PSC into Podemos) are applied when parties are compared across elections. A 2024

winner is term-limited in 2025–2028 if he or she was re-elected; this classification agrees with an independent proxy in the companion files in 94% of the municipalities.

S4.3. What the data show

The status quo is described in Section 5.1 of the article; this section adds the data used in the calibration. The share of the planned value of special transfers with technical impediments, which anchors the capacity gradient (Section S4.6), is 23% in municipalities with fewer than 5,000 inhabitants, 19% in those with 5,000–10,000, 14% with 10,000–20,000, 12% with 20,000–50,000, 10% with 50,000–100,000 and 16% above 100,000. The mayoral elections anchor the municipal module (Section S4.5). In Santa Catarina's 2024 mayoral elections, 71% of the eligible incumbents ran and 67% of those who ran won. Moreover, 35% of the re-elected mayors had changed party since 2020, and 62% of those who changed (18 of 29) joined PL, the governor's party. Of the open seats, those in which the incumbent did not run, 18% stayed with the outgoing mayor's party (15% in 2020), while none of the 40 seats whose incumbent ran and lost in 2024 did, nor any such seat in 2020. Finally, PL's share of the state's mayors rose from 9% in 2020 to 31% in 2024.

S4.4. Parameters

Table S4 lists the 71 parameters with their uniform priors, their groups (used in the sensitivity analysis of Supplementary Material 06) and their sources. Following Kydland and Prescott (1996) and Hansen and Heckman (1996), three kinds of quantities are distinguished. Quantities measured directly, such as populations, revenues, party affiliations, seats and the status-quo allocation, are data (Sections S4.1–S4.3). Parameters of the current regime are calibrated to observed moments: the municipal electoral parameters to the 2016–2024 elections in Santa Catarina (Section S4.5), the completion of transfer-funded works to the TCU audits and the elasticity of completion to capacity to the impediment gradient (Section S4.6), and diversion to the audit-based estimates of Ferraz and Finan (2011) (Section S4.7). Behavioral parameters of institutions that do not exist yet, such as list-makers' weights, the excludability of regional benefits or the value of reappointment to a conde, are given wide priors, and the results are reported as functions of them.

Table S4. Parameters, priors and sources

Parameter	Group	Description	Prior (uniform)	Source or anchor
s0	Chamber	status-quo share of deputies' effort on brokerage	[0.5, 0.8]	premise P1; unmeasured
pi	Chamber	list-makers' weight on brokerage relative to voters'	[0.2, 1.2]	assumption
kappa	Chamber	list-makers' weight on legislation relative to voters'	[0.5, 2]	assumption
t0	Chamber	voters' weight on brokerage skill relative to legislative skill (open list)	[0.5, 2]	assumption
rho_LB	Chamber	correlation between legislative and brokerage skill	[−0.3, 0.5]	assumption
sig_eps	Chamber	noise in personal votes, relative to the weight on legislative skill	[1, 3]	assumption
sig_nu	Chamber	list-makers' idiosyncratic criteria (loyalty, faction)	[0.5, 3]	assumption; Jones et al. (2002)
k_asp	Chamber	serious aspirants per seat	[2, 5]	assumption
gam_L	Chamber	legislative productivity per SD of legislative skill	[0.1, 0.3]	assumption
eta_X	Relocation	share of abolished amendment money re-created in discretionary channels	[0, 1]	Rapporteur-amendment (RP9) episode 2022-23: half to individual amendments, half to ministries; committee amendments R$0.3 bn (2022) to R$6.9 bn (2023) and R$14.2 bn (2024) (Ronconi et al. 2025)
chi_c	Relocation	responsiveness of a co-partisan conde to deputies' requests	[0, 1]	assumption
s_bar	Projects	cost of candidate regional projects / county money of a term	[0.3, 1]	assumption
sig_bar	Projects	mean net scale economies of regional projects	[−0.1, 0.3]	Bel & Warner (2015); Bel & Sebő (2021)
sd_sig	Projects	dispersion of scale economies across projects	[0.1, 0.3]	assumption

Parameter	Group	Description	Prior (uniform)	Source or anchor
p_polo	Projects	probability that a project is hosted by the county's largest city	[0.4, 0.8]	assumption
D0	Projects	distance decay of project benefits (km)	[30, 90]	assumption
c_host	Projects	host's local cost (land, permits, opposition), share of project cost	[0, 0.1]	assumption
kap_cred	Projects	conde's private credit per real of regional investment	[0, 0.3]	assumption; Litschig (2012) credit logic
q_loc0	Implementation	completion of transfer-funded municipal works within a term	[0.55, 0.8]	TCU (2019): 37.5% paralyzed or unfinished; TCU (2023): 41% (SC 38.9%)
nu	Implementation	elasticity of completion to technical capacity	[0, 0.3]	2025 special transfers to SC: planned value with technical impediments 23% (<5,000 inh.) to 10-12% (20,000-100,000); implied elasticity 0.13-0.17
phi_cx	Implementation	completion penalty of regional (complex) projects	[0, 0.3]	assumption
g_pool	Implementation	share of the capacity gap closed by a county technical team	[0, 0.5]	assumption
alpha_e	Implementation	proportional gain in completion at full supervision effort	[0.1, 0.5]	assumption
c_e	Implementation	conde's cost of full effort (share of term budget)	[0.02, 0.2]	assumption
v_inc	Implementation	salvage value of an abandoned work per real	[0.1, 0.5]	assumption
a_ab	Implementation	abandonment of unfinished projects when the conde's party changes	[0.2, 0.6]	assumption; disruption after turnover (Akhtari et al. 2022)
a_same	Implementation	abandonment when the conde changes within the same party	[0, 0.2]	assumption
beta_d	Implementation	value of a work completed one term late, per real of on-time value	[0.7, 0.9]	assumption
c_adm	Implementation	county administration, share of county money	[0.01, 0.04]	assumption

Parameter	Group	Description	Prior (uniform)	Source or anchor
alpha_u	Implementation	value gain of completed projects from unmeasured quality effort	[0, 0.2]	assumption; multitask (Holmstrom & Milgrom 1991)
alpha_a	Implementation	completion gain per SD of the conde's competence	[0, 0.15]	assumption
omega_h	Cooperation	completion lost if the host obstructs	[0.2, 0.6]	assumption
nu_c	Cooperation	elasticity of completion to the share of the project's cost funded	[1, 3]	assumption; TCU: insufficient counterpart funds
eps_x	Cooperation	excludability of benefits / enforcement against non-payers	[0, 0.5]	assumption; Law 11.107/2005 (exclusion of defaulting members)
zeta_opp	Cooperation	discount on benefits credited to a rival party's conde	[0, 0.4]	assumption; alignment and adoption (Macinko et al. 2010)
s_shock	Cooperation	scale of private cost shocks relative to the mayor's stake	[0.25, 1]	assumption
ell	Cooperation	mayors' localism premium	[0, 0.5]	Gerber & Gibson (2009); Ten Cate (2023)
gam	Cooperation	credit for regional projects accruing to participating mayors	[0, 0.3]	assumption
beta_y	Cooperation	annual discount factor	[0.85, 0.95]	assumption
theta_lo	Conde	lower bound of condes' weight on value (theta ~ U[theta_lo, 1])	[0.2, 0.7]	assumption
V_R	Conde	value of reappointment to the conde, share of term budget	[0, 0.3]	assumption
rho_R	Conde	retrospective weight of mayors' gains in the college vote (logit)	[0, 3]	assumption; Ferraz & Finan (2008) for voters
phi_P	Conde	party loyalty in the college vote (logit)	[1.5, 4]	Novaes (2018): 31-39% of mayors switch parties
kap_party	Conde	share of a conde's commitments honored by a co-partisan successor	[0.3, 1]	assumption
c_T	Conde	conde's cost per real moved in the pre-election tilt	[0, 0.5]	assumption

Parameter	Group	Description	Prior (uniform)	Source or anchor
alpha_V	Conde	electoral return to local money (logit per unit of revenue)	[0.5, 6]	Litschig & Morrison (2013): +10 points of incumbent-party re-election (base 0.17) for 20% more spending; about 3.5 logit points per unit (conversion in Section S4.8; 95% interval about 0.1-5.9)
h_V	Conde	scale of diminishing electoral returns (share of revenue)	[0.02, 0.1]	assumption
r_run	Municipal	eligible incumbents who run	[0.69, 0.708]	SC 2020: 0.690; 2024: 0.708
p_win0	Municipal	incumbents who run and win	[0.669, 0.742]	SC 2024: 0.669; 2020: 0.742
h_sw	Municipal	switching hazard of running incumbents	[0.18, 0.45]	calibrated: re-elected mayors who changed party, SC 2020: 0.136; 2024: 0.349
beta_bw	Municipal	bandwagon toward the conde's party (switchers and open seats)	[0, 0.3]	assumption; 62% of 2024 switchers joined the governor's party
r_keep	Municipal	open seats kept by the outgoing mayor's party	[0.148, 0.185]	SC 2020: 0.148; 2024: 0.185 (incumbent not running)
c_dir	Municipal	precision of the statewide partisan climate at each election (Dirichlet)	[10, 40]	calibrated: change in mayors' party shares, SC 2016-20: 0.15; 2020-24: 0.23
f_SC	Diversion	Santa Catarina factor on municipal diversion rates	[0.5, 1]	Ferraz & Finan (2011): 0.055 eligible, 0.074 term-limited
d_c0	Diversion	baseline diversion and capture in county procurement	[0.04, 0.12]	Ferraz & Finan (2011); lower part of the 5-23% model-implied corruption tax on firms in Colonnelli & Prem (2022)
D_max	Diversion	discipline from a contestable reappointment	[0.15, 0.5]	Ferraz & Finan (2011): 27-36%
g_ten	Diversion	rise in capture per extra term of party tenure	[0, 0.3]	assumption
w_cap	Targets	share of capacity in the county score	[0.3, 0.6]	assumption
w_noise	Targets	share of noise in the county score	[0.2, 0.5]	Kane & Staiger (2002)
phi_c	Targets	share of counties failing at baseline	[0.1, 0.3]	design (open in the proposal)
s_T	Targets	share of next-term money withheld from a failing county	[0.2, 0.6]	design (proposal: loss of access to new resources)

Parameter	Group	Description	Prior (uniform)	Source or anchor
Xi	Targets	Union's resolve to enforce territorial sanctions	[1.5, 3]	assumption
varrho	Targets	need gradient of the harm from withholding money	[0.5, 1.5]	assumption
p_I	LRG	efficacy of inspection (share of attempts blocked)	[0, 0.3]	Olken (2007); Avis et al. (2018)
s_D	LRG	penalty per real when diversion is detected	[0, 2]	assumption
g_TA	LRG	completion gain from technical assistance at zero capacity	[0, 0.3]	Lautharte et al. (2021); companion paper
c_A	LRG	cost of commissioners per real overseen	[0, 0.03]	companion paper: R$350k per year
zeta	Need	inequality aversion in need-weighted value	[0, 1]	assumption
u_contract	Design	share of non-excludable benefit made enforceable by contracts (R3)	[0.5, 0.9]	assumption; Kresch (2020)
gam_cb	Design	credit shared with participating mayors under co-branding (R4)	[0.3, 0.7]	assumption
e_cap	Design	share of relocation blocked by a cap on discretionary channels (R13)	[0.3, 0.9]	assumption

Note. All priors are uniform, and worlds are drawn by a scrambled Sobol' sequence. "Assumption" marks priors without a direct empirical anchor.

S4.5. The municipal module

The municipal module determines how the colleges of mayors change between conde terms, so its electoral parameters, except the bandwagon toward the conde's party, which is given a prior (Table S4), are calibrated to the two observed cycles (2016–2020 and 2020–2024). The run, win and retention rates are set to ranges spanned by the two cycles. The switching hazard, which counts draws of a new affiliation that may return the mayor's own party, and the precision of the statewide partisan climate are chosen so that two moments fall within the observed values: the share of re-elected mayors who changed party (0.14 and 0.35) and the change in the party composition of the mayors between elections, measured by the total variation distance between the party shares (0.15 and 0.23). Table S5 and Figure S1 compare the moments of the simulated status quo, over the three simulated municipal elections, with the data. The simulated moments were not used to set the priors of any behavioral parameter of the new institutions.

Table S5. The municipal module: simulated status quo and observed moments

Moment	Model, status quo (median [5th, 95th])	Santa Catarina (two cycles)
Eligible incumbents who run	0.70 [0.67, 0.73]	0.69; 0.71
Runners who win	0.70 [0.66, 0.75]	0.74; 0.67
Re-elected who changed party	0.25 [0.15, 0.36]	0.14; 0.35
Open seats kept by the party	0.17 [0.13, 0.20]	0.15; 0.18
Change in party shares (TV)	0.16 [0.11, 0.24]	0.15; 0.23
Term-limited share after an election	0.33 [0.31, 0.35]	0.41; 0.32

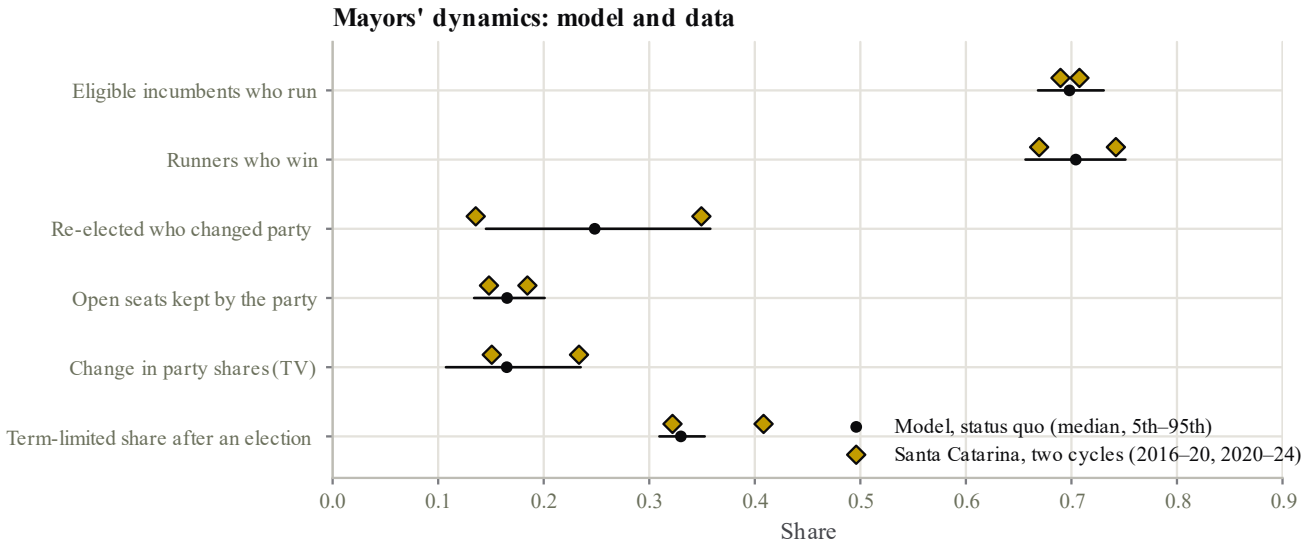


Figure S1. Calibration of the municipal module: moments of the simulated status quo (distribution across worlds and elections) against the two observed cycles in Santa Catarina.

S4.6. Completion and the capacity gradient

The completion of transfer-funded municipal works within a term, $\bar{q} \in [0.55,0.80]$, brackets the shares of federally funded works that the TCU found paralyzed or unfinished: 37.5% in 2019 and 41% in 2023, and 38.9% in Santa Catarina (TCU, 2019, 2023). The elasticity of completion to capacity, ν, is anchored on the 2025 special transfers. Reading the share of planned value with technical impediments as a proxy for non-completion, a least-squares fit of the logarithm of the share without impediments on the logarithm of mean capacity across the six size classes, weighted by the number of plans in each class, gives an elasticity of 0.13. Without the largest class, whose higher share of impediments may reflect the complexity of its projects rather than its capacity, the elasticity is 0.17. The prior is $U[0,0.3]$, whose midpoint, 0.15, lies between the two estimates.

S4.7. Diversion, discipline and capture

Municipal diversion follows Ferraz and Finan (2011): audited federal money diverted by mayors who can be re-elected (5.5%) and by term-limited mayors (7.4%), scaled by a factor between 0.5 and 1 because corruption in Santa Catarina is likely lower than the national average. Their estimates also imply that re-election incentives reduce the share of audited resources diverted by about a quarter to a third of the level of term-limited mayors (27% in their text; their estimate with full controls, 0.027, is 36% of the term-limited mean of 0.074), and that the difference vanishes when the mayor's party holds about 55% of the municipal council. These magnitudes are used for the discipline of a contestable reappointment, $D_{max} \in [0.15,0.5]$, and for the share of the college at which it vanishes. Capture in county procurement, $d_{c0} \in [0.04,0.12]$, is informed by the same magnitudes and by the lower part of the 5%–23% average corruption tax on firms implied by the effects of randomized audits (Colonnelli & Prem, 2022).

S4.8. The electoral return to money

Litschig and Morrison (2013) find that about 20% more local spending over four years raised the probability that the incumbent party was re-elected by about 10 percentage points, against a comparison mean of 0.17, which is a semi-elasticity of about 0.5. The model's win probability is a logit, so the semi-elasticity must be converted into logit points. At a base of 0.17, a semi-elasticity of 0.5 corresponds to about $0.5/(0.17 \times 0.83) \approx 3.5$ logit points per unit of revenue. The 95% confidence interval of their most precise estimate, (0.002, 0.167), that is, 0.2 to 16.7 percentage points for the 20% increase, corresponds to about 0.1–5.9 logit points. The prior for α_V is $U[0.5,6]$.

S4.9. Relocation

The share of abolished money re-created in discretionary channels, η, is drawn from $U[0,1]$. The 2022–2023 episode (Section 2.3 of the article) suggests the upper half: after the Supreme Court abolished rapporteur amendments, their money was divided between individual amendments and ministries' regular appropriations, and committee amendments grew from R$ 0.3 billion in 2022 to R$ 14.2 billion in 2024 (Congresso em Foco, 2022; Ronconi et al., 2025). The prior is nevertheless not narrowed, because the abolition of all individual amendments, which the proposal envisages, has no precedent, and the results are reported by ranges of η.

Supplementary Material 05 – Additional Results and Robustness Checks

This unit reports results that the article summarizes. Section S5.1 gives the main outcomes of every design and compares each design with the proposal as stated in the same worlds. Section S5.2 maps who gains and who loses, Section S5.3 extends the per-capita rule to the federation and Section S5.4 reports the robustness checks.

S5.1. All designs

Table S6 reports the main outcomes of every design, and Table S7 the paired difference between each design and the proposal as stated (R0) in the same worlds. Two features of these tables deserve attention. First, several designs change no outcome other than the one they target. Population-weighted colleges change who controls the condes (Supplementary Material 07, Section S7.1) but not the value of the money; the bonus for aligned condes moves money between counties without changing cooperation; and the cap on discretionary channels raises the Chamber's legislative effort (Figure 3b of the article) and real value while lowering need-weighted value, because the relocated money keeps a geography that targets need better than a per-capita split. Second, the designs that raise cooperation (contracts, co-branding, no co-financing, the technical conde and the robust variant) raise completion and the regional share together. This is why, among the designs without the managerial law, their paired gains in real value over the proposal are the largest (Table S7). One caveat applies to the cooperation statistic of R5. Without co-financing, participants have nothing to pay, so the statistic is close to one by construction, and its change in Table S7 is not comparable with that of the other designs.

Table S6. Main outcomes of every design

Design	Real value (%)	Need-weighted value (%)	Regional share (%)	Completion (%)	Cooperation	Losing municipalities (%)	Worlds: real value up	Worlds: need-weighted up	Worlds: better than R0 (need-weighted)
CL Closed lists only	0.0	0.0	–	–	–	0	–	–	–
L0 Status quo + managerial law	2.2	2.3	–	–	–	0	85%	86%	89%
D0 Per-capita transfers, no counties	0.6	−3.0	–	–	–	56	99%	7%	49%
R0 Proposal as stated	0.6	−2.3	10	72	0.60	56	70%	18%	–
R1 One-term conde	0.4	−2.6	9	69	0.54	56	65%	14%	20%
R2 No target bar on reappointment	0.6	−2.3	10	71	0.61	56	70%	18%	41%
R3 Binding consortium contracts	1.1	−1.7	13	80	0.86	56	80%	25%	88%
R4 Co-branding of projects	1.5	−1.5	18	78	0.74	56	82%	28%	90%
R5 No municipal co-financing	1.6	−1.4	22	83	0.99	56	85%	29%	92%
R6 Population-weighted college	0.6	−2.3	10	73	0.61	56	71%	18%	46%
R7 Dual majority for the plan	0.5	−2.5	9	71	0.60	56	69%	16%	32%
R8 Peer-relative targets	0.6	−2.3	10	72	0.60	56	71%	18%	71%
R9 Sanction on the conde only	0.6	−2.3	10	72	0.60	56	70%	18%	77%
R10 Need-based split	0.5	−1.7	10	72	0.60	55	69%	22%	95%
R11 Bonus to aligned condes	0.6	−2.3	10	72	0.60	56	70%	18%	47%
R12 Technical conde	2.0	−1.0	24	75	0.74	56	85%	33%	91%
R13 Cap on discretionary channels	1.4	−5.3	10	72	0.60	56	70%	18%	18%
R14 Proposal + managerial law	3.2	−0.6	11	74	0.62	54	87%	45%	85%
R15 Robust variant	4.6	0.8	22	85	0.97	54	93%	57%	94%
R16 Robust variant + need-based split	4.6	1.4	22	85	0.97	54	93%	64%	96%

Note. Medians across 2,000 worlds. Value changes are relative to the status quo in the same world. Losing municipalities are those that receive less value than in the status quo.

Table S7. Each design compared with the proposal as stated, in the same worlds

Design	Δ real value (pp)	Δ need-weighted value (pp)	Δ cooperation	Δ completion
R1 One-term conde	−0.17 (22%)	−0.17 (20%)	−0.05 (5%)	−0.02 (30%)
R2 No target bar on reappointment	0.00 (42%)	0.00 (41%)	0.00 (59%)	0.00 (36%)
R3 Binding consortium contracts	+0.41 (87%)	+0.36 (88%)	+0.22 (100%)	+0.06 (88%)
R4 Co-branding of projects	+0.68 (91%)	+0.60 (90%)	+0.13 (100%)	+0.05 (82%)
R5 No municipal co-financing	+0.73 (92%)	+0.65 (92%)	+0.39 (100%)	+0.10 (93%)
R6 Population-weighted college	0.00 (51%)	−0.01 (46%)	0.00 (53%)	0.00 (53%)
R7 Dual majority for the plan	−0.05 (32%)	−0.04 (32%)	0.00 (28%)	0.00 (33%)
R8 Peer-relative targets	0.00 (50%)	+0.02 (71%)	0.00 (60%)	0.00 (51%)
R9 Sanction on the conde only	−0.01 (30%)	+0.02 (77%)	0.00 (1%)	0.00 (46%)
R10 Need-based split	−0.06 (22%)	+0.67 (95%)	0.00 (6%)	0.00 (34%)
R11 Bonus to aligned condes	0.00 (46%)	0.00 (47%)	0.00 (3%)	0.00 (46%)
R12 Technical conde	+1.08 (92%)	+0.95 (91%)	+0.13 (99%)	+0.03 (69%)
R13 Cap on discretionary channels	+0.31 (70%)	−1.26 (18%)	0.00 (23%)	0.00 (15%)
R14 Proposal + managerial law	+2.11 (84%)	+2.11 (85%)	+0.01 (86%)	+0.01 (67%)
R15 Robust variant	+3.71 (94%)	+3.62 (94%)	+0.34 (100%)	+0.12 (94%)
R16 Robust variant + need-based split	+3.69 (93%)	+4.52 (96%)	+0.34 (100%)	+0.12 (93%)

Note. Median paired difference with R0 in the same world; in parentheses, the share of worlds in which the design is above R0.

S5.2. Who gains and who loses

Figure S2 maps the median change in the value that each municipality receives, in reais per inhabitant per year. The losses are concentrated in the small municipalities that receive the most amendments today, across the whole state, and the gains in the larger municipalities. The robust variant with a need-based split shifts the pattern only slightly, because need varies more within counties than between them: 74% of the population-weighted variance of need is within counties.

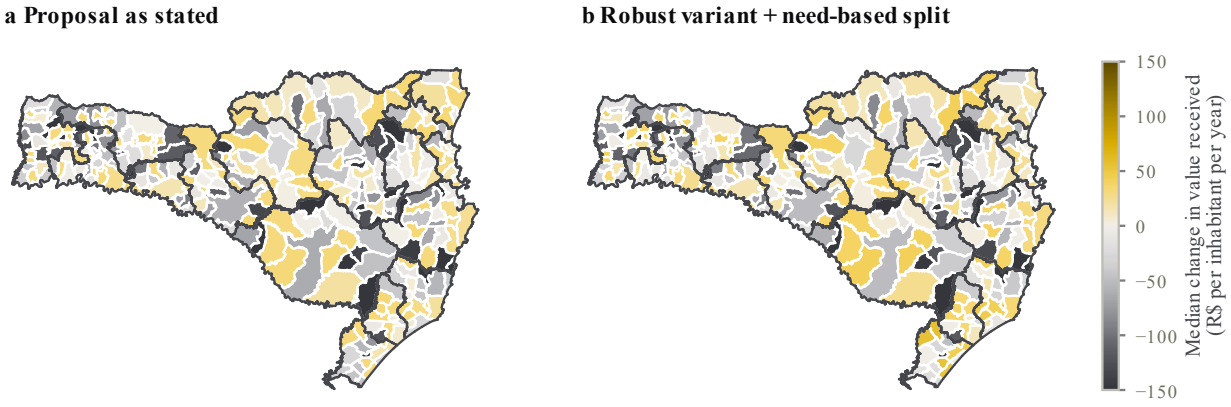


Figure S2. Who gains and who loses: median change in the value received by each municipality relative to the status quo (R$ per inhabitant per year), under (a) the proposal as stated (R0) and (b) the robust variant with a need-based split between counties (R16). Lines are county borders.

S5.3. National accounting

Section 6.9 of the article asks what the per-capita rule would do if it were applied to the whole federation. Figure S3 answers by distributing the individual amendments of all deputies among the states in proportion to population. The national per-capita amount would be R$ 94 per inhabitant per year. States such as Roraima, Amapá and Acre would lose more than 70% of their deputies' amendments, and São Paulo would gain 60%. If the rule were applied to deputies' and senators' amendments together, 18 states, with 203 deputies and 54 senators, would lose.

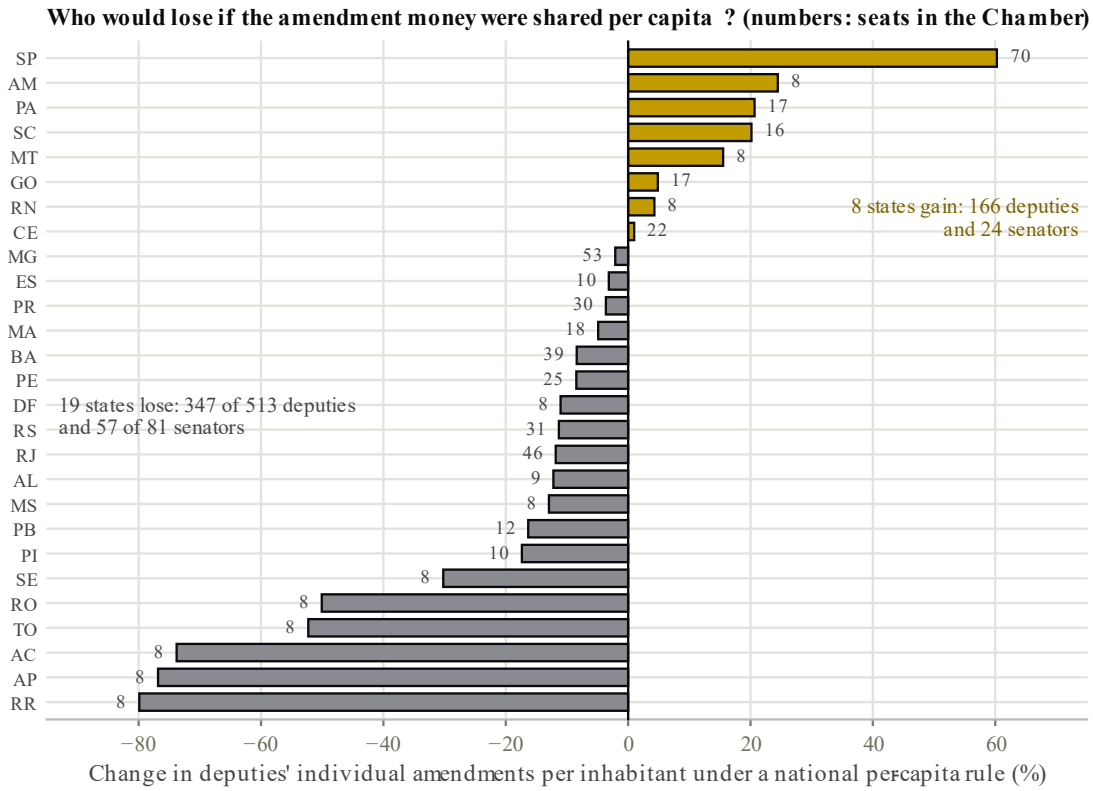


Figure S3. National accounting: change in deputies' individual amendments per inhabitant if they were distributed among the states in proportion to population (2025 quotas; 2022 Census population); the numbers are each state's seats in the Chamber.

S5.4. Robustness checks

Table S8 reports the main results under the alternative specifications that Section 6.10 of the article summarizes. Each variant is run over 1,000 worlds with the parameter draws of the first 1,000 worlds of the main experiment and its own random numbers, and the main specification is re-run in the same way for comparison. The status-quo variants are described in Supplementary Material 04 (Section S4.2).

The status-quo variants change the results that depend on the geography of today's money. Giving the three deputies who used special transfers little the pooled geography of the other thirteen, or giving every deputy that pooled geography, changes the proposal's need-weighted result little (−2.5% and −2.4%, against −2.4% in the main specification over the same worlds), although the largest municipalities then gain more (112% and 149% for those above 100,000 inhabitants, against 76%), because the pooled geography gives them less today. With the geography of the deputies elected in 2026, in which the transfers of the seven newcomers are predicted from their 2026 votes, today's amendments go less to the smallest municipalities and more to poor municipalities of 5,000–20,000 inhabitants, so they are more need-targeted: the neediest fifth of the population receives 2.4 times its population share of the money, against 2.0. The proposal then lowers need-weighted value by 3.5%, raising it in 13% of the worlds, and the robust variant lowers it too (−0.4%, positive in 46%). Because this variant predicts the geography of seven of the sixteen deputies from a regression whose coefficient on GDP

per capita is imprecise, it is reported as an indication of how much the comparison depends on who sits in the Chamber. If half of each quota, the health half, were allocated by population today, the losses from the per-capita rule would be smaller: the proposal would change real value by 0.3% and need-weighted value by −1.0%, and the robust variant would raise need-weighted value by 2.0%, in 71% of the worlds.

The other checks change the value results less. Counties formed by the 8 health macro-regions change them little; with them, PL elects 49% of the condes in 2027 on average, and the same party holds a county for all three terms in 26% of the counties. Party-line voting in the colleges raises PL's share of condes to 53% and persistence to 28% without changing the value results, and the strong-bandwagon variant raises persistence to 49% of the counties while leaving the value results and PL's share of the condes almost unchanged. Without any capacity gradient in completion ($\nu = 0$), the per-capita reallocation adds no real value, and the proposal changes real value by −0.03% and need-weighted value by −3.2%.

Two of these variants need further detail. The first is the strong-bandwagon variant, which draws β_{bw} from $U[0.3,0.6]$, above the main prior $U[0,0.3]$, and sets the conde's private cost of the pre-election tilt to zero; as a result, the money moved toward co-partisan mayors rises from 0.2% to 6.5% of all the money, on average across worlds.

The second is the variant with targets on outcome indicators. In this variant, the weight of the conde's delivery in the county score, w_p, is drawn from $U[0.003,0.033]$ and that of noise from $U[0.02,0.20]$, by rescaling the draws of w_c and w_n, and capacity takes the rest of the variance. The range of w_p matches the share of the within-state variance of outcome indicators that the companion paper attributes to the component that changes with each administration, between 0 and 3.3% (Companion paper, 2026); in the main specification, by comparison, w_p is between about 0.08 and 0.5. The run includes R0, R2 and R8, and the main specification is re-run with the same three designs, so that the effects of the bar (R0 against R2) and of absolute targets (R0 against R8) can be compared in the same worlds. With outcome targets, the bar raises the conde's supervision effort in 70% of the worlds (85% with the targets of the main specification), by a median of 0.006 (0.024); it lowers cooperation in 72% (60%) and raises real value in 49% (55%). Absolute targets fail the poorest third of the counties more often than peer comparison does in 96% of the worlds (86%), by a median of 22 percentage points (11), and lower need-weighted value in 71% (70%). Table S8 reports the failure rates.

Table S8. Main results under alternative specifications

Variant	Design	Real value (%)	Need-weighted value (%)	Worlds: need-weighted up	Cooperation	Regional share (%)	PL condes, 2027; 2035 (mean)	Counties held by one party for three terms (mean)	Evaluations failed, poorest; richest third (mean)
Main specification (1,000 worlds)	D0	0.5	−3.0	7%	–	–	–	–	–
Main specification (1,000 worlds)	R0	0.7	−2.4	19%	0.60	10	45%; 42%	20%	36%; 9%
Main specification (1,000 worlds)	R8	0.7	−2.3	19%	0.60	10	45%; 42%	20%	23%; 23%
Main specification (1,000 worlds)	R15	4.7	0.9	59%	0.97	22	45%; 42%	21%	24%; 24%
Status quo: pooled Pix geography	D0	0.7	−3.0	8%	–	–	–	–	–
Status quo: pooled Pix geography	R0	0.9	−2.4	19%	0.60	10	45%; 42%	20%	36%; 9%
Status quo: pooled Pix geography	R15	5.0	1.0	59%	0.97	22	45%; 42%	21%	24%; 24%
Status quo: three low users with the pooled geography	D0	0.6	−3.2	8%	–	–	–	–	–
Status quo: three low users with the pooled geography	R0	0.8	−2.5	18%	0.60	10	45%; 42%	20%	36%; 9%
Status quo: three low users with the pooled geography	R15	4.9	0.7	57%	0.97	22	45%; 42%	21%	24%; 24%
Status quo: half of each quota by population (health)	D0	0.3	−1.5	7%	–	–	–	–	–
Status quo: half of each quota by population (health)	R0	0.3	−1.0	28%	0.60	10	45%; 42%	20%	36%; 9%
Status quo: half of each quota by population (health)	R15	4.1	2.0	71%	0.97	22	45%; 42%	21%	24%; 24%
Status quo: the deputies elected in 2026	D0	0.5	−4.3	5%	–	–	–	–	–
Status quo: the deputies elected in 2026	R0	0.6	−3.5	13%	0.60	10	45%; 42%	20%	36%; 9%
Status quo: the deputies elected in 2026	R15	4.5	−0.4	46%	0.97	22	45%; 42%	21%	24%; 24%
Counties: 8 health macro-regions	D0	0.5	−3.0	7%	–	–	–	–	–
Counties: 8 health macro-regions	R0	0.5	−2.5	16%	0.58	9	49%; 48%	26%	34%; 12%
Counties: 8 health macro-regions	R15	4.6	0.7	57%	0.95	21	49%; 48%	28%	23%; 25%
Party-line college voting	D0	0.5	−3.0	7%	–	–	–	–	–
Party-line college voting	R0	0.7	−2.2	19%	0.62	11	53%; 51%	28%	36%; 9%

Variant	Design	Real value (%)	Need-weighted value (%)	Worlds: need-weighted up	Cooperation	Regional share (%)	PL condes, 2027; 2035 (mean)	Counties held by one party for three terms (mean)	Evaluations failed, poorest; richest third (mean)
Party-line college voting	R15	4.7	0.9	59%	0.97	22	53%; 51%	28%	24%; 24%
Bandwagon above the prior (β_{bw} from U[0.3,0.6]), costless tilting	R0	0.6	−2.2	19%	0.61	10	45%; 43%	49%	36%; 9%
Bandwagon above the prior (β_{bw} from U[0.3,0.6]), costless tilting	R6	0.7	−2.4	18%	0.61	10	42%; 29%	16%	37%; 9%
No capacity gradient in completion ($\nu = 0$)	D0	0.0	−3.7	0%	–	–	–	–	–
No capacity gradient in completion ($\nu = 0$)	R0	0.0	−3.2	11%	0.58	8	45%; 42%	19%	33%; 10%
No capacity gradient in completion ($\nu = 0$)	R15	3.8	0.1	51%	0.96	20	45%; 42%	20%	23%; 25%
Targets on outcome indicators	R0	0.6	−2.4	19%	0.60	10	45%; 42%	19%	42%; 1%
Targets on outcome indicators	R8	0.6	−2.3	19%	0.60	10	45%; 42%	19%	18%; 30%

Note. Each variant is run over 1,000 worlds with the parameter draws of the first 1,000 worlds of the main experiment and its own random numbers; the main specification is re-run in the same way for comparison (its R8 row comes from a run of R0, R2 and R8 with the same draws and random numbers). Medians across worlds unless stated otherwise. With targets on outcome indicators, the conde's delivery accounts for 0.3%–3.3% of the variance of the county score and noise for 2%–20%.

Supplementary Material 06 – Sensitivity Analysis and Scenario Discovery

Most results of the article depend on parameters of institutions that do not exist yet, and this unit explains how the parameters that matter were identified. The analysis has two steps, which answer different questions. The first step, in Section S6.1, asks which parameters drive each outcome: it screens all parameters and decomposes the variance of the main outcomes by groups of parameters. The second step, in Section S6.2, asks in which regions of the parameter space a result holds: it compares the worlds in which a result holds with the others and summarizes the difference in a few thresholds.

S6.1. Screening and variance decomposition

All 71 parameters were first screened with the elementary-effects method (Morris, 1991), with ten trajectories. Each trajectory uses four levels placed at the midpoints of four equal intervals of each prior (1/8, 3/8, 5/8 and 7/8 of the range), in steps of two levels, so that no evaluation falls on a boundary of a prior. The reason for avoiding the boundaries is substantive: at complete relocation, $\eta = 1$, for example, there is no county money, and the county outcomes are zero by construction. For each parameter, the screening computes the mean absolute elementary effect on each outcome of the proposal as stated (Figure S4).

The second part of the analysis computes first-order and total-effect Sobol' indices for groups of parameters, using the estimators of Saltelli et al. (2010) and Jansen (1999) with 1,024 base samples per matrix and bootstrap confidence intervals. The proposal requires 12,288 model runs. The robust variant requires 14,336, because its analysis also varies the parameters of the managerial law and of its design elements. Table S9 reports the total-effect indices; the first-order indices and the confidence intervals are in the replication files.

The total-effect index needs a careful reading. The total-effect index of a group is the expected share of the variance of the output that remains when all the other parameters are fixed. It therefore includes the group's interactions with the others, so the indices of one output can sum to more than one, and an index is not the share of the variance that the group explains on its own.

Table S9. Grouped total-effect Sobol' indices

Output	Chamber	Reloca-tion	Projects	Implemen-tation	Coopera-tion	Conde	Municipal	Diversion	Targets	Need	LRG	Design
Need-weighted value	0	0.34	0.27	0.21	0.04	0.01	0	0.01	0	0.44	–	–
Real value	0	0.30	0.56	0.44	0.09	0.02	0	0.03	0	0	–	–
Cooperation	0	0	0.30	0.30	0.44	0.05	0.02	0.02	0	0	–	–
Regional share	0	0	0.64	0.32	0.13	0.04	0	0.02	0	0	–	–
Completion	0	0	0.36	0.72	0.25	0.18	0.06	0.07	0.02	0	–	–
Legislative effort	0.63	0.40	0	0	0	0	0	0	0	0	–	–
Party concentration (2035)	0	0.02	0.05	0.04	0.04	0.50	0.34	0.01	0.01	0	–	–
Persistence of failure	0	0	0.51	0.53	0.41	0.27	0.10	0.14	0.43	0	–	–
R15: need-weighted value	0	0.25	0.38	0.23	0.01	0	0	0.01	0	0.33	0.10	0
R15: real value	0	0.31	0.51	0.30	0.02	0.01	0	0.02	0	0	0.12	0.01
R15: cooperation	0	0	0.23	0.34	0.76	0.11	0.01	0	0	0	0.11	0.16
R15: regional share	0	0	0.73	0.27	0.08	0.01	0	0	0	0	0	0.02
R15: completion	0	0	0.20	0.86	0.17	0.07	0.02	0.02	0.01	0	0.06	0.05

Note. Total-effect indices (Jansen estimator) for the proposal as stated (R0) relative to the status quo, N = 1024 base samples per matrix, 12,288 model runs; bootstrap 95% intervals in the replication files. Rows marked R15 refer to the robust variant, whose sensitivity analysis also varies the parameters of the managerial law (LRG) and of its design elements; N = 1024, 14,336 runs.

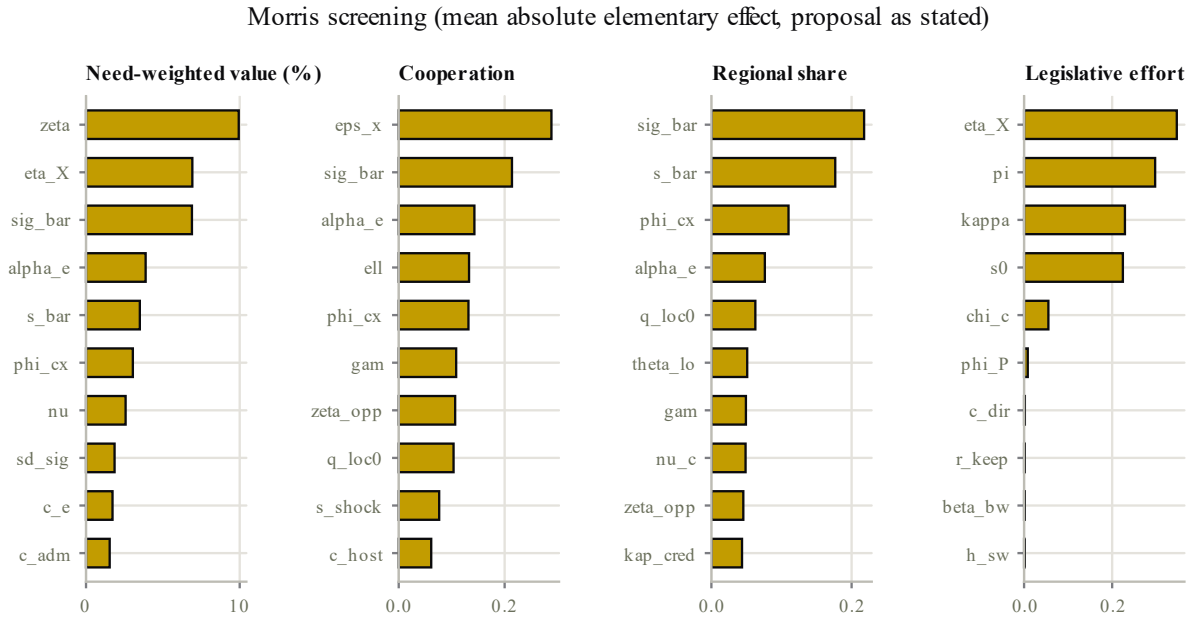


Figure S4. Morris screening: the ten parameters with the largest mean absolute elementary effects on four outcomes of the proposal as stated.

S6.2. Scenario discovery

For each binary result, scenario discovery reports three results: the share of worlds in which the result holds; the parameters whose distributions differ most between the worlds in which it holds and the others, measured by the two-sample Kolmogorov–Smirnov statistic, D; and a classification tree of depth three with at least 60 worlds per leaf (Saltelli et al., 2008; Bryant & Lempert, 2010). The trees are in out/cc2_discovery.json of the replication package. The main splits, from which the article takes its rounded thresholds, are the following.

- *The proposal raises need-weighted value* (18% of the worlds). The parameters that separate these worlds are the weight on need ζ ($D = 0.45$), the mean scale economies $\bar{\sigma}$ (0.39), the cost of the menu $\bar{s}$ (0.23) and the completion penalty ϕ_{cx} (0.21). The tree's positive leaf is $\zeta \leq 0.308$, $\bar{\sigma} > 0.080$ and a completion of transfer-funded works $\bar{q} > 0.679$, a region that makes up 8% of the worlds and in which the proposal raises need-weighted value in 77% of them; 34% of the worlds in which it does so lie in this region. Without the condition on $\bar{q}$, the region covers 17% of the worlds, and the proposal raises need-weighted value in 59% of them.
- *The proposal raises real value* (70%). The separating parameters are $\bar{\sigma}$ (0.35), the capacity elasticity ν (0.31), ϕ_{cx} (0.25) and the cost of county administration (0.24). The negative leaves have $\bar{\sigma} \leq 0.083$ with $\nu \leq 0.137$ and $\phi_{cx} > 0.109$, or $\bar{\sigma} > 0.083$ with $\phi_{cx} > 0.177$ and $\nu \leq 0.068$.
- *The county institutions add real value* (R0 above D0, 52%). The separating parameters are $\bar{\sigma}$ (0.43), ϕ_{cx} (0.28), the dispersion of scale economies across projects (0.18) and $\bar{s}$ (0.18). There are three positive leaves: $\bar{\sigma} > 0.092$ with $\phi_{cx} \leq 0.177$ (31% of the worlds, positive in 85% of them); $\bar{\sigma} > 0.092$ with $\phi_{cx} > 0.177$ and productive supervision ($\alpha_e > 0.225$); and $\bar{\sigma} \leq 0.092$ with $\phi_{cx} \leq 0.170$ and widely

dispersed scale economies (standard deviation above 0.227), where some candidate projects are worth funding even though the mean is low.

- *Cooperation at least 0.75* (15%) or *at least 0.8* (8%). The separating parameters are excludability ϵ_x ($D = 0.50$ and 0.56), localism ℓ (0.25 and 0.29), $\bar{\sigma}$ (0.27 and 0.26) and the completion of transfer-funded works $\bar{q}$ (0.23 and 0.26). For 0.75, the positive leaf is $\epsilon_x > 0.307$, $\ell \leq 0.207$ and $\phi_{cx} \leq 0.113$ (6% of the worlds; cooperation reaches 0.75 in 67% of them); with the first two conditions only, the region covers 16% of the worlds, and cooperation reaches 0.75 in 48% of them. For 0.8, the positive leaf is $\epsilon_x > 0.401$, $\ell \leq 0.153$ and a third split on the discipline parameter, $D_{max} \leq 0.322$ (3% of the worlds; 68%). This third split has no evident mechanism and is not interpreted; without it, the leaf covers 6% of the worlds, of which 48% reach 0.8, against 6% of the other worlds.
- *Regional share at least 20%* (23%). The separating parameters are $\bar{\sigma}$ (0.48), $\bar{s}$ (0.32) and ϕ_{cx} (0.30).
- *Legislative effort rises for every party* (95%). The separating parameters are list-makers' weight on legislation κ (0.72), their weight on brokerage π (0.53) and relocation η (0.47). The exceptions make up 5% of the worlds, and 74% of them have $\kappa \leq 0.832$, $\pi > 0.782$ and $\eta > 0.507$.
- *The bar raises real value* (57%) and *the bar raises completion* (59%). No parameter separates these worlds well ($D \leq 0.12$). The trees, with an accuracy of 0.62 and 0.64 against 0.57 and 0.59 for the majority class, split first on the value of reappointment to the conde, V_R.
- *The robust variant raises need-weighted value* (57%). The separating parameters are ζ (0.37), $\bar{\sigma}$ (0.28), η (0.28), inspection p_I (0.18) and technical assistance g_{TA} (0.18). The positive leaves are $\zeta \leq 0.506$ with either $\bar{\sigma} > 0.128$ or inspection that blocks more than 10.1% of attempts, and $\zeta > 0.506$ with $\eta > 0.670$ and $g_{TA} > 0.119$ (10% of the worlds, positive in 84% of them). In the last leaf, most of the money keeps the status-quo geography, and technical assistance raises completion where capacity is lowest.

Supplementary Material 07 – Colleges of Mayors and the Electoral Connection

This unit describes two sets of real political data from Santa Catarina on which the article relies. Section S7.1 describes the colleges of mayors elected in 2024, which elect the first condes in the model and determine who controls them (Section 6.2 of the article). Section S7.2 describes the electoral connection between deputies' votes and their transfers, which bears on premise P1 and supplies the targeting regression used to complete the status quo (Section 5.1 of the article).

S7.1. The real colleges of mayors

Table S10 describes the 17 colleges of mayors elected in 2024 and the plurality winner of each college under one vote per mayor and under population weights. The table shows how the plurality rule turns small leads into control. In Serra Catarinense, for example, PL wins under party-line voting with one vote per mayor with 22% of the mayors, and in several counties two parties tie.

The voting rule also determines how much power the largest municipality of each county has. Figure S5 reports the normalized Banzhaf power (Banzhaf, 1965; Felsenthal & Machover, 1998) of each county's largest municipality under three voting rules. With one vote per mayor, the largest municipality's power is between 0.03 and 0.17, below its population share in every county. With population weights, by contrast, it decides alone in five counties (power of one) and has more power than its population share in all of them. A dual majority of mayors and population gives it intermediate power.

Table S10. The colleges of mayors elected in 2024, by county

#	County (health region)	Municipalities	Largest city (share of population)	Plurality, one mayor one vote	Plurality, population-weighted	Banzhaf power of largest city: one vote / dual / population
1	Extremo Oeste	30	São Miguel do Oeste (18%)	PL, 37% of mayors	PL, 49% of population	0.033 / 0.09 / 0.22
2	Oeste	27	Chapecó (62%)	MDB, 48% of mayors	PSD, 66% of population	0.037 / 0.22 / 1.00
3	Xanxerê	21	Xanxerê (25%)	MDB / PL (tie), 24% of mayors	MDB, 31% of population	0.048 / 0.14 / 0.31
4	Alto Uruguai Catarinense	13	Concórdia (53%)	PL, 54% of mayors	PL, 69% of population	0.077 / 0.31 / 1.00
5	Meio Oeste	20	Campos Novos (19%)	MDB / PSDB (tie), 25% of mayors	PSDB, 34% of population	0.050 / 0.10 / 0.21
6	Alto Vale do Rio do Peixe	20	Caçador (26%)	MDB, 35% of mayors	PL, 43% of population	0.050 / 0.12 / 0.30
7	Planalto Norte	13	São Bento do Sul (22%)	PL, 46% of mayors	PL, 68% of population	0.077 / 0.15 / 0.25

#	County (health region)	Municipalities	Largest city (share of population)	Plurality, one mayor one vote	Plurality, population-weighted	Banzhaf power of largest city: one vote / dual / population
8	Serra Catarinense	18	Lages (56%)	PL, 22% of mayors	Cidadania, 57% of population	0.056 / 0.23 / 1.00
9	Alto Vale do Itajaí	28	Rio do Sul (23%)	PL, 43% of mayors	PL, 53% of population	0.036 / 0.12 / 0.35
10	Extremo Sul Catarinense	15	Araranguá (32%)	PP, 40% of mayors	MDB, 57% of population	0.067 / 0.22 / 0.51
11	Carbonífera	12	Criciúma (48%)	PP, 42% of mayors	PSD, 62% of population	0.083 / 0.27 / 0.98
12	Médio Vale do Itajaí	14	Blumenau (44%)	PL, 57% of mayors	PL, 92% of population	0.071 / 0.23 / 0.80
13	Vale do Itapocu	7	Jaraguá do Sul (55%)	PSD / PL (tie), 29% of mayors	MDB, 55% of population	0.143 / 0.41 / 1.00
14	Laguna	18	Tubarão (29%)	PL, 39% of mayors	PL, 66% of population	0.056 / 0.15 / 0.40
15	Grande Florianópolis	22	Florianópolis (40%)	PL, 45% of mayors	PSD, 60% of population	0.045 / 0.16 / 0.48
16	Nordeste	6	Joinville (79%)	PSD / MDB (tie), 33% of mayors	Novo, 79% of population	0.167 / 0.35 / 1.00
17	Foz do Rio Itajaí	11	Itajaí (33%)	PSD / MDB (tie), 36% of mayors	PL, 46% of population	0.091 / 0.22 / 0.43

Note. Plurality winners assume party-line voting. Banzhaf power is normalized to sum to one across the college's members.

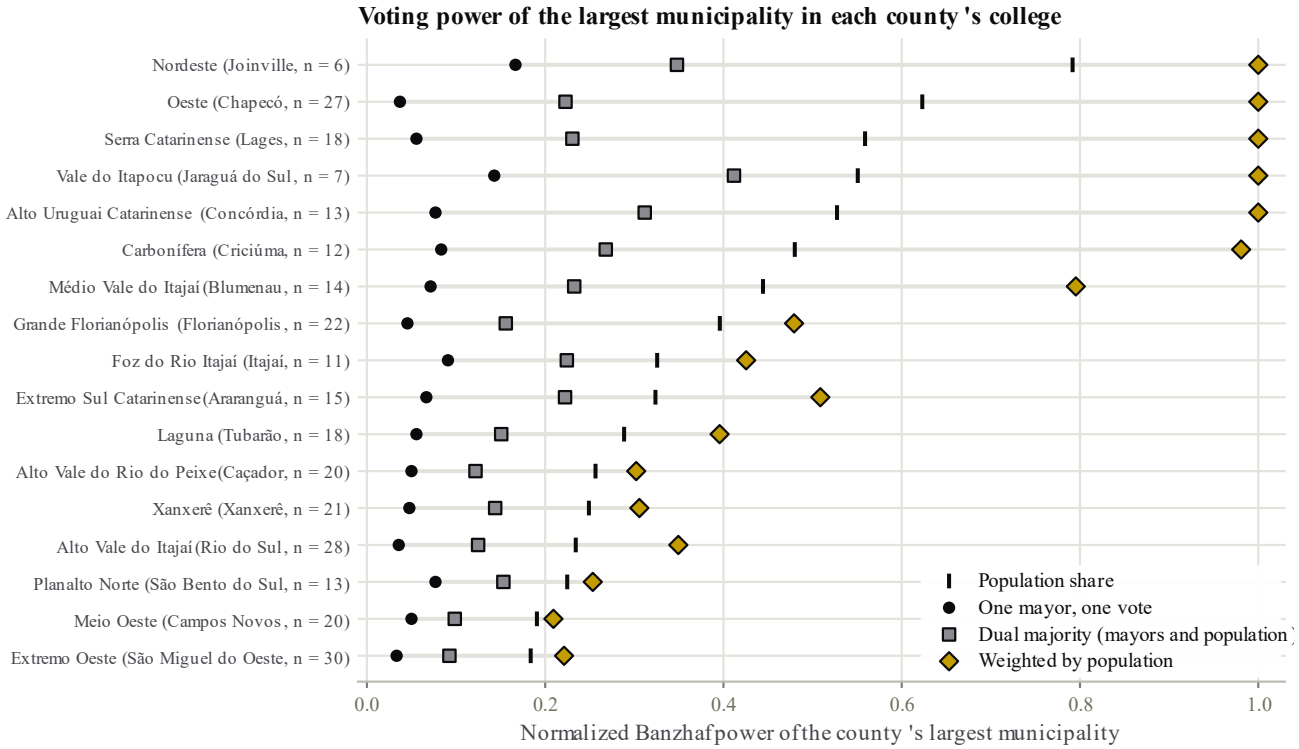


Figure S5. Voting power of the largest municipality of each county: normalized Banzhaf power under one vote per mayor, a dual majority of mayors and population, and population-weighted votes, compared with the municipality's population share.

S7.2. The electoral connection

S7.2.1. Data and matching

This section uses the votes by candidate, municipality and zone in the 2022 and 2026 general elections in Santa Catarina (Tribunal Superior Eleitoral, 2026), summed over the zones of each municipality and matched to the statistical municipal codes with the crosswalk of the companion files (Supplementary Material 04, Section S4.2). The 16 deputies elected in 2022 are matched by ballot name to the authors of the 2025 special transfers, and the 2026 candidates are matched to the 2022 deputies by full name. Vote shares are shares of the nominal votes for the Chamber in the municipality.

S7.2.2. Where the votes come from

The median deputy elected in 2022 drew 15% of his or her votes from a single municipality (between 7% and 38% across deputies), and half of them from 15 municipalities (between 6 and 28). The vote bases are mostly urban: the municipalities that supplied half of a deputy's votes hold 31% of the state's population (median across deputies), but they received 22% of the deputy's special transfers.

S7.2.3. Where the transfers go

Within the state, a deputy's money follows his or her votes once municipal size is taken into account. The evidence comes from a Poisson pseudo-maximum-likelihood regression of the transfers planned by each of the 13 deputies with more than R$ 10 million in special transfers to each municipality (3,835 deputy-municipality pairs). The regressors are the logarithm of the deputy's share of the municipality's 2022 vote, the logarithms of population and GDP per capita and a mayor of the deputy's party, with deputy fixed effects and standard errors clustered by municipality. The regression gives an elasticity of 0.69 (standard error 0.07) to the vote share, 0.45 (0.08) to population and −0.57 (0.30) to GDP per capita, and a factor of 1.9 for a co-partisan mayor (coefficient 0.65, 0.19; Figure S6a). The median deputy funds 29 municipalities.

The fitted model also serves to complete the status quo. Applied to the 2022 votes of the three deputies who used special transfers little, and restricted to the 29 municipalities with the largest predicted transfers of each, it gives their geography in the status quo (Supplementary Material 04, Section S4.2). The prediction is smoother than real allocations: across the thirteen, the median correlation between the predicted shares, restricted to 29 municipalities, and the actual ones is 0.28, and those 29 municipalities received a median of 43% of each deputy's money. The same model, applied to the 2026 votes of the seven deputies elected for the first time in 2026, gives the status-quo variant for the Chamber elected in 2026 (Table S8 in Supplementary Material 05).

S7.2.4. Transfers and votes in 2026

Of the 16 deputies elected in 2022, 13 ran again for the Chamber in 2026 and 9 were re-elected; three of the 13 ran for another party, and one of the other three was elected to the Senate. For the 13, a regression of the deputy's share of each municipality's 2026 vote (in percentage points) on the deputy's 2025 transfers to the municipality (in R$ 100 per inhabitant), the deputy's 2022 share there and a co-partisan mayor, with deputy and municipality fixed effects (3,835 pairs; standard errors clustered by municipality), gives a coefficient of 2.9 (0.5) on transfers and 1.4 (0.3) on a co-partisan mayor (Figure S6b).

The coefficient on transfers remains positive in alternative specifications. It is 1.9 (0.5) for the change in the share without the 2022 share as a control, 3.5 (0.6) without the three deputies who changed party, 2.9 (0.5) without the three who used special transfers little, 3.8 (0.6) when municipalities are weighted by their 2026 votes, and 0.44 (0.07) in the logarithm of votes. If the association were causal, the 2025 transfers of these deputies would account for about 34,000 of their votes in 2026, 1.9% of the total, or about R$ 5,700 of 2025 transfers per vote. Moreover, if the amendments of other years follow similar geographies, the coefficient also captures their association with votes, and the money per vote of all the amendments would be larger.

These are associations and not causal effects, because deputies may send money where they expect to gain, and the mayors and local allies who receive it may campaign for them. What they show is that amendment

money in Santa Catarina is targeted to deputies' electoral bases and co-partisan mayors and moves with their votes, which is the brokerage that the proposal seeks to remove from the Chamber.

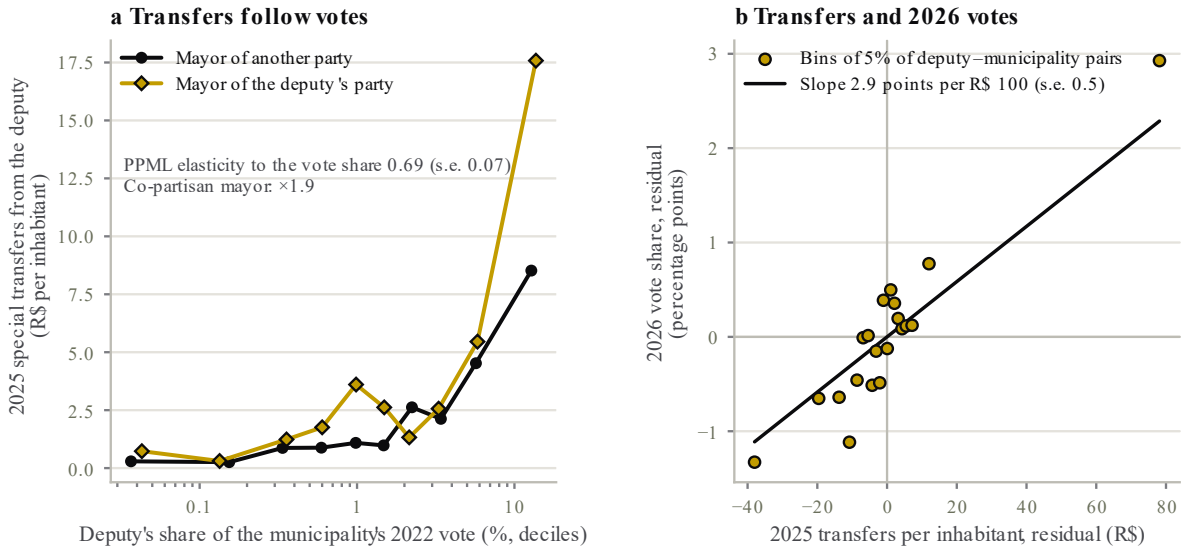


Figure S6. The electoral connection in Santa Catarina. (a) 2025 special transfers per inhabitant from the 13 deputies with more than R$ 10 million in special transfers, by deciles of the deputy's share of the municipality's 2022 vote, for municipalities with a mayor of the deputy's party and of another party; the text gives the PPML estimates. (b) Deputy's share of the municipality's 2026 vote against the deputy's 2025 transfers per inhabitant, both net of deputy and municipality fixed effects, of the deputy's 2022 share and of a co-partisan mayor, in 20 bins of equal size, with the fitted slope. Sources: votes for federal deputy, Tribunal Superior Eleitoral (2026); 2025 special transfers, Transferegov, via Transparência Brasil and the companion package.

Supplementary Material 08 – Tests and Replication

This unit describes the tests of the engine (Section S8.1) and the replication package (Section S8.2).

S8.1. Tests

Before the experiment, the engine passed three tests (code/cc2_tests_engine.py), each of which checks a property that the model must have by construction. First, every design allocates exactly the same money as the status quo over the twelve years. Second, with perfect inputs (no diversion, full completion, no scale economies, no administration and no quality effort), the value delivered by the status quo and by the proposal without co-financing equals the money. Third, when all the abolished money relocates ($\eta = 1$), the proposal, with either college and with a technical conde, reproduces the status quo exactly, municipality by municipality. The propositions were checked as reported in Supplementary Material 03 (Section S3.10 and Table S2).

S8.2. Replication

The replication package (folder Replication_Package of the Supplementary Material) contains the code, the derived data, the companion data used and all outputs. The TSE files of votes by candidate, municipality and zone are not redistributed, because of their size; code/prep_tse.py rebuilds the derived vote files from them when they are placed in data/raw_tse/, and otherwise uses the derived files included in the package.

The script run_all.sh rebuilds every number, table and figure of the article and of this Supplementary Material. It runs, in order, the data preparation, the tests, the propositions, the main experiment (2,000 worlds and 21 designs, about seven minutes on the test machine used for this study), the instrumented run of 300 worlds used in Figures 6a, 7a and 7c and in the mechanism statistics, the analysis, the robustness checks, the sensitivity analyses, the figures and the tables; the whole pipeline takes about 20 minutes. Every random draw is seeded, and a second, clean run of the pipeline in a fresh copy of the package reproduced every output.

References

Abrucio, F. L., Filippim, E. S., & Dieguez, R. C. (2013). Inovação na cooperação intermunicipal no Brasil: A experiência da Federação Catarinense de Municípios (Fecam) na construção de consórcios públicos. *Revista de Administração Pública, 47*(6), 1543–1568. https://doi.org/10.1590/S0034-76122013000600010

Agência IBGE Notícias. (2020, December 2). *Munic: Em 2019, número de servidores municipais cresce apenas 0,1% frente a 2018*. Instituto Brasileiro de Geografia e Estatística. https://agenciadenoticias.ibge.gov.br/agencia-sala-de-imprensa/29564-munic-em-2019-numero-de-servidores-municipais-cresce-apenas-0-1-frente-a-2018.html

Agência Senado. (2025, December 1). *O que são emendas parlamentares e como são definidas?* https://www12.senado.leg.br/noticias/materias/2025/12/01/o-que-sao-emendas-parlamentares-e-como-sao-definidas

Akhtari, M., Moreira, D., & Trucco, L. (2022). Political turnover, bureaucratic turnover, and the quality of public services. *American Economic Review, 112*(2), 442–493. https://doi.org/10.1257/aer.20171867

Alston, L. J., & Mueller, B. (2006). Pork for policy: Executive and legislative exchange in Brazil. *The Journal of Law, Economics, and Organization, 22*(1), 87–114. https://doi.org/10.1093/jleo/ewj001

Ames, B. (1995). Electoral strategy under open-list proportional representation. *American Journal of Political Science, 39*(2), 406–433. https://doi.org/10.2307/2111619

Avis, E., Ferraz, C., & Finan, F. (2018). Do government audits reduce corruption? Estimating the impacts of exposing corrupt politicians. *Journal of Political Economy, 126*(5), 1912–1964. https://doi.org/10.1086/699209

Avis, E., Ferraz, C., Finan, F., & Varjão, C. (2022). Money and politics: The effects of campaign spending limits on political entry and competition. *American Economic Journal: Applied Economics, 14*(4), 167–199. https://doi.org/10.1257/app.20200296

Baião, A. L., & Couto, C. G. (2017). A eficácia do pork barrel: A importância de emendas orçamentárias e prefeitos aliados na eleição de deputados. *Opinião Pública, 23*(3), 714–753. https://doi.org/10.1590/1807-01912017233714

Banzhaf, J. F., III. (1965). Weighted voting doesn't work: A mathematical analysis. *Rutgers Law Review, 19*(2), 317–343.

Baron, D. P., & Ferejohn, J. A. (1989). Bargaining in legislatures. *American Political Science Review, 83*(4), 1181–1206. https://doi.org/10.2307/1961664

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy, 76*(2), 169–217. https://doi.org/10.1086/259394

Bel, G., & Sebő, M. (2021). Does inter-municipal cooperation really reduce delivery costs? An empirical evaluation of the role of scale economies, transaction costs, and governance arrangements. *Urban Affairs Review, 57*(1), 153–188. https://doi.org/10.1177/1078087419839492

Bel, G., & Warner, M. E. (2015). Inter-municipal cooperation and costs: Expectations and evidence. *Public Administration, 93*(1), 52–67. https://doi.org/10.1111/padm.12104

Bertholini, F., Pereira, C., & Rennó, L. (2018). Pork is policy: Dissipative inclusion at the local level. *Governance, 31*(4), 701–720. https://doi.org/10.1111/gove.12331

Brollo, F., & Nannicini, T. (2012). Tying your enemy's hands in close races: The politics of federal transfers in Brazil. *American Political Science Review, 106*(4), 742–761. https://doi.org/10.1017/S0003055412000433

Bryant, B. P., & Lempert, R. J. (2010). Thinking inside the box: A participatory, computer-assisted approach to scenario discovery. *Technological Forecasting and Social Change*, *77*(1), 34–49. https://doi.org/10.1016/j.techfore.2009.08.002

Colonnelli, E., & Prem, M. (2022). Corruption and firms. *The Review of Economic Studies, 89*(2), 695–732. https://doi.org/10.1093/restud/rdab040

Comissão Intergestores Bipartite de Santa Catarina. (2024). *Deliberação 38/CIB/2024: Macrorregiões de saúde do estado de Santa Catarina* [Approved 7 March 2024; rectified 4 June 2024]. Secretaria de Estado da Saúde de Santa Catarina. https://saude.sc.gov.br/index.php/pt/legislacao/legislacao-geral/deliberacoes/deliberacoes-2021/038-07-03-macrorregioes-de-saude-do-estado-de-santa-catarina-retificada-em-04-06-2024/download

Companion paper. (2026). *Can performance replace votes? An ex-ante evaluation of Brazil's proposed Lei de Responsabilidade Gerencial* [Working paper]. [Author and repository to be completed.]

Congresso em Foco. (2022, December 22). *Congresso vota orçamento após excluir Bolsa Família do teto de gastos*. https://www.congressoemfoco.com.br/noticia/44389/congresso-vota-orcamento-apos-excluir-bolsa-familia-do-teto-de-gastos

Conselho Nacional de Secretários de Saúde. (2023, March 16). [Slide presentation on planning, health-care networks and regionalization, Jornada SES/CTs]. https://www.conass.org.br/wp-content/uploads/2023/03/Apresentacao-Jornada-SES-CTs-Planej-Redes-Regionalizacao_marco23-1.pdf

Decreto-Lei n.º 27/2020, de 17 de junho. (2020). *Diário da República*, 1.ª série, n.º 116, 3–6. https://diariodarepublica.pt/dr/detalhe/decreto-lei/27-2020-135951161

Feiock, R. C. (2013). The institutional collective action framework. *Policy Studies Journal, 41*(3), 397–425. https://doi.org/10.1111/psj.12023

Felsenthal, D. S., & Machover, M. (1998). *The measurement of voting power: Theory and practice, problems and paradoxes*. Edward Elgar.

Ferraz, C., & Finan, F. (2008). Exposing corrupt politicians: The effects of Brazil's publicly released audits on electoral outcomes. *The Quarterly Journal of Economics, 123*(2), 703–745. https://doi.org/10.1162/qjec.2008.123.2.703

Ferraz, C., & Finan, F. (2011). Electoral accountability and corruption: Evidence from the audits of local governments. *American Economic Review, 101*(4), 1274–1311. https://doi.org/10.1257/aer.101.4.1274

Filippim, E. S., & Abrucio, F. L. (2010). Quando descentralizar é concentrar poder: O papel do governo estadual na experiência catarinense. *Revista de Administração Contemporânea, 14*(2), 212–228. https://doi.org/10.1590/S1415-65552010000200003

Firpo, S., Ponczek, V., & Sanfelice, V. (2015). The relationship between federal budget amendments and local electoral power. *Journal of Development Economics, 116*, 186–198. https://doi.org/10.1016/j.jdeveco.2015.04.005

Gazeta do Povo. (2024, August 14). *Dino suspende emendas impositivas até que Congresso crie regras de transparência*. https://www.gazetadopovo.com.br/economia/dino-suspende-emendas-impositivas-ate-que-congresso-crie-regras-transparencia/

Gerber, E. R., & Gibson, C. C. (2009). Balancing regionalism and localism: How institutions and incentives shape American transportation policy. *American Journal of Political Science, 53*(3), 633–648. https://doi.org/10.1111/j.1540-5907.2009.00391.x

Groseclose, T., & Snyder, J. M., Jr. (1996). Buying supermajorities. *American Political Science Review, 90*(2), 303–315. https://doi.org/10.2307/2082886

Hansen, L. P., & Heckman, J. J. (1996). The empirical foundations of calibration. *Journal of Economic Perspectives, 10*(1), 87–104. https://doi.org/10.1257/jep.10.1.87

Holmstrom, B., & Milgrom, P. (1991). Multitask principal–agent analyses: Incentive contracts, asset ownership, and job design. *Journal of Law, Economics, & Organization, 7*(Special issue), 24–52. https://doi.org/10.1093/jleo/7.special_issue.24

InfoMoney. (2022, December 19). *STF declara "orçamento secreto" inconstitucional, por 6 votos a 5*. https://www.infomoney.com.br/?p=2011420

Jansen, M. J. W. (1999). Analysis of variance designs for model output. *Computer Physics Communications, 117*(1–2), 35–43. https://doi.org/10.1016/S0010-4655(98)00154-4

Jones, M. P., Saiegh, S., Spiller, P. T., & Tommasi, M. (2002). Amateur legislators–professional politicians: The consequences of party-centered electoral rules in a federal system. *American Journal of Political Science, 46*(3), 656–669. https://doi.org/10.2307/3088406

Kane, T. J., & Staiger, D. O. (2002). The promise and pitfalls of using imprecise school accountability measures. *Journal of Economic Perspectives, 16*(4), 91–114. https://doi.org/10.1257/089533002320950993

Kornai, J., Maskin, E., & Roland, G. (2003). Understanding the soft budget constraint. *Journal of Economic Literature, 41*(4), 1095–1136. https://doi.org/10.1257/002205103771799999

Kresch, E. P. (2020). The buck stops where? Federalism, uncertainty, and investment in the Brazilian water and sanitation sector. *American Economic Journal: Economic Policy, 12*(3), 374–401. https://doi.org/10.1257/pol.20180327

Kydland, F. E., & Prescott, E. C. (1996). The computational experiment: An econometric tool. *Journal of Economic Perspectives, 10*(1), 69–85. https://doi.org/10.1257/jep.10.1.69

Lautharte, I., Oliveira, V. H. de, & Loureiro, A. (2021). *Incentives for mayors to improve learning: Evidence from state reforms in Ceará, Brazil* (Policy Research Working Paper No. 9509). World Bank. https://doi.org/10.1596/1813-9450-9509

Lei Complementar nº 210, de 25 de novembro de 2024. (2024). *Diário Oficial da União*, 26 November 2024. https://www.planalto.gov.br/ccivil_03/leis/lcp/lcp210.htm

Lei nº 11.107, de 6 de abril de 2005. (2005). *Diário Oficial da União*, 7 April 2005. https://www.planalto.gov.br/ccivil_03/_ato2004-2006/2005/lei/l11107.htm

Lei n.º 37/2020, de 17 de agosto. (2020). *Diário da República*, 1.ª série, n.º 159, 3–5. https://diariodarepublica.pt/dr/detalhe/lei/37-2020-140431101

Limongi, F., & Figueiredo, A. (2005). Processo orçamentário e comportamento legislativo: Emendas individuais, apoio ao Executivo e programas de governo. *Dados, 48*(4), 737–776. https://doi.org/10.1590/S0011-52582005000400002

Litschig, S. (2012). Are rules-based government programs shielded from special-interest politics? Evidence from revenue-sharing transfers in Brazil. *Journal of Public Economics, 96*(11–12), 1047–1060. https://doi.org/10.1016/j.jpubeco.2012.08.010

Litschig, S., & Morrison, K. M. (2013). The impact of intergovernmental transfers on education outcomes and poverty reduction. *American Economic Journal: Applied Economics, 5*(4), 206–240. https://doi.org/10.1257/app.5.4.206

Loi organique n° 2017-1338 du 15 septembre 2017 pour la confiance dans la vie politique. (2017). *Journal officiel de la République française*. https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000035567936

Macinko, J., Dourado, I., Aquino, R., Bonolo, P. de F., Lima-Costa, M. F., Medina, M. G., Mota, E., de Oliveira, V. B., & Turci, M. A. (2010). Major expansion of primary care in Brazil linked to decline in unnecessary hospitalization. *Health Affairs, 29*(12), 2149–2160. https://doi.org/10.1377/hlthaff.2010.0251

Mendonça, C. (2020, October 14). CCDR: Dos cinco novos presidentes, quatro foram os únicos candidatos à eleição. *Postal*. https://postal.pt/politica/ccdr-dos-cinco-novos-presidentes-quatro-foram-os-unicos-candidatos-a-eleicao/

Morris, M. D. (1991). Factorial sampling plans for preliminary computational experiments. *Technometrics, 33*(2), 161–174. https://doi.org/10.1080/00401706.1991.10484804

Novaes, L. M. (2018). Disloyal brokers and weak parties. *American Journal of Political Science, 62*(1), 84–98. https://doi.org/10.1111/ajps.12331

Olken, B. A. (2007). Monitoring corruption: Evidence from a field experiment in Indonesia. *Journal of Political Economy, 115*(2), 200–249. https://doi.org/10.1086/517935

Pavan, B. (2026, August 30). Emendas parlamentares saltam 315% em 11 anos e mostram orçamento sob controle do Legislativo, aponta estudo. *Times Brasil*. https://timesbrasil.com.br/brasil/emendas-parlamentares-saltam-315-em-11-anos-e-mostram-orcamento-sob-controle-do-legislativo-aponta-estudo/

Pereira, C., & Mueller, B. (2004). The cost of governing: Strategic behavior of the president and legislators in Brazil's budgetary process. *Comparative Political Studies, 37*(7), 781–815. https://doi.org/10.1177/0010414004266866

Raile, E. D., Pereira, C., & Power, T. J. (2011). The executive toolbox: Building legislative support in a multiparty presidential regime. *Political Research Quarterly, 64*(2), 323–334. https://doi.org/10.1177/1065912909355711

Romer, T., & Rosenthal, H. (1978). Political resource allocation, controlled agendas, and the status quo. *Public Choice, 33*(4), 27–43. https://doi.org/10.1007/BF03187594

Ronconi, G. B. A., Fialho, G. F., & Greggianin, E. (2025). *Caráter nacional e individualização das emendas de comissão* (Estudo No. 18). Câmara dos Deputados, Consultoria de Orçamento e Fiscalização Financeira. https://www2.camara.leg.br/orcamento-da-uniao/estudos/2025/estudo-emendas-de-comissao-versao-formatada

Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., & Tarantola, S. (2010). Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. *Computer Physics Communications, 181*(2), 259–270. https://doi.org/10.1016/j.cpc.2009.09.018

Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., & Tarantola, S. (2008). *Global sensitivity analysis: The primer*. John Wiley & Sons. https://doi.org/10.1002/9780470725184

Samuels, D. (2003). *Ambition, federalism, and legislative politics in Brazil*. Cambridge University Press. https://doi.org/10.1017/CBO9780511510366

Samuels, D. J. (2002). Pork barreling is not credit claiming or advertising: Campaign finance and the sources of the personal vote in Brazil. *The Journal of Politics, 64*(3), 845–863. https://doi.org/10.1111/0022-3816.00149

Ten Cate, L. (2023). Quem recebe e que diferença fazem: Emendas orçamentárias nos municípios brasileiros pós EC 86/2015 (2015–2019). *Opinião Pública, 29*(3), 792–826. https://doi.org/10.1590/1807-01912023293792

Tollini, H., & Mendes, M. (2024). *É assim em todo lugar? Emendas parlamentares no Brasil e em 11 países da OCDE*. Insper. https://static.poder360.com.br/2025/01/e-assim-em-todo-lugar-emendas-parlamentares-brasil-ocde-insper-2024.pdf

Tribunal de Contas da União. (2019). *Auditoria operacional sobre obras paralisadas* (Acórdão 1.079/2019 – Plenário, TC 011.196/2018-1). https://portal.tcu.gov.br/data/files/8F/12/9D/F2/4281671023455957E18818A8/Auditoria_operacional_obras_paralisadas.pdf

Tribunal de Contas da União. (2023, October 18). *Brasil tem 8,6 mil obras paralisadas, financiadas com recursos federais* (Acórdão 2.134/2023 – Plenário) [News release]. https://portal.tcu.gov.br/imprensa/noticias/brasil-tem-86-mil-obras-paralisadas-financiadas-com-recursos-federais

Tribunal Superior Eleitoral. (2026). *Votação nominal por candidato, município e zona: Eleições gerais de 2022 e 2026, Santa Catarina* [Data set; files generated on 8 October 2026]. Portal de Dados Abertos do TSE. https://dadosabertos.tse.jus.br